

Remote Work and City Structure[†]

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Relative to remote work, working downtown facilitates valuable interactions with other in-office workers but entails commuting costs. The resulting coordination mechanism can lead to multiple stationary equilibria with varying levels of remote work. Temporary reductions in commuters, as in the COVID-19 pandemic, can lead to persistently large fractions of remote workers. Cell phone-based mobility data for the United States shows commuting trips in the largest cities, which are more likely to exhibit multiplicity, have stabilized at only 60 percent of prepandemic levels, while they are fully back in smaller cities. Cities with permanently low commuting experience average welfare losses of 2.3 percent. (JEL I12, J22, M54, R23, R31)

Many of us work at least partially from home these days. A phenomenon that was only marginal a few years ago has become widespread and has challenged, in potentially profound ways, the basic organization of work in firms and cities. Information technology has improved to a point where workers can talk, chat, and send information in documents, databases, audio, and video without delays and at extremely low costs. However, although information technology clearly complements remote work, significant advances in information and communication technologies (ICT) increased remote work only marginally between 1980 and 2019. In contrast, since the COVID-19 pandemic, remote work has boomed. Suddenly, commuting has declined by half in many of our largest cities and seems to have stabilized at those low levels.¹ What generated this abrupt and seemingly permanent increase in work from home since the pandemic? What are its consequences for the structure of cities? Will the change be uniform across them? Importantly, what are the welfare consequences of permanent shifts in commuting and agglomeration patterns? In this paper, we aim to address these questions by proposing, and estimating for most US cities, a dynamic theory of city structure with a labor delivery choice. We also

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¹These trends have been reported from survey data (Barrero, Bloom, and Davis 2023; Brynjolfsson et al. 2020; Aksoy et al. 2022; Brynjolfsson et al. 2023), administrative data (Dalton, Dey, and Loewenstein 2022), changes in job advertisements for remote positions (Hansen et al. 2023), or changes in internet usage of firms (Matthies and Kwan 2022).

contrast and verify its implications with data on individual cell phone mobility and zip code-level house prices.

At the core of our theory is the idea that the value of working at the central business district (CBD)—in short, “office work”—depends on proximity to other workers. Workers’ productivity is enhanced by interactions with others in their office, across businesses, and more generally with other workers at city centers. This is an old idea from Marshall, of course. Remote work offers workers and employers the possibility to work in the same city but without interacting in person. Since workers do not internalize many of the benefits they convey to others, this alternative possibility leads to a novel coordination problem: Workers prefer to work in the CBD only if others do too, but they prefer to work from home otherwise. In a dynamic economy, where workers face idiosyncratic preferences for remote work (e.g., having a baby or remodeling) and fixed costs from switching between labor delivery modes (e.g., setting up a home office or buying a car), this mechanism can lead to multiple stationary equilibria with different permanent shares of commuters. In these cases, abrupt changes to the number of commuters can make a city converge to a stationary equilibrium that is very different from the initial one. For many places, we postulate, this abrupt change was the COVID-19 pandemic and the lockdowns and self-isolation it generated. This one-time temporary change has led, for several US cities, to a permanent change in one of the most enduring characteristics of human organization: work at city centers.

Of course, not all cities are equal. Some cities specialize in industries where interactions between workers lead to large spillovers and correspondingly large populations. Other cities specialize in industries with low spillovers and are small. Commuting costs can vary dramatically depending on transport infrastructure and geographic characteristics. Furthermore, the ability to work from home varies sharply with the worker’s occupation, and occupations are, clearly, not uniformly distributed in space. We all know about the concentration of bankers in New York or programmers in Palo Alto. All these characteristics of cities determine the extent to which the coordination problem we identify leads to multiplicity in stationary equilibria. We characterize these conditions and prove that, naturally, the intensity of the coordination problems, and therefore the likelihood of multiple stationary equilibria, increases with the strength of agglomeration forces minus that of congestion forces and in the relative productivity of remote work to office work. We show that, in the United States, the particular configuration of urban characteristics that leads to multiplicity in stationary equilibria was prevalent before the pandemic in many of the larger cities. That is, we estimate parameter configurations for large cities that, in general, although not always, put them in the “multiplicity cone”: the parameter set where cities exhibit multiplicity in stationary equilibria.

Our finding that large cities tend to be in the multiplicity cone implies that the abrupt reduction in commuters caused by the COVID-19 pandemic made many of them switch to a path that seems to be converging to the low-commuting stationary equilibrium. Of course, even in the stationary equilibrium with little commuting, some people commute; their idiosyncratic preference for office work and the fixed cost of switching make it the better choice for them. Our interpretation is that the pandemic reduced the share of commuters below the share of commuters in the low-commuting stationary equilibrium. Hence, commuting has increased in some of

these cities, but only slightly, and has stabilized way below prepandemic levels. In contrast, small cities tend to have parameter configurations outside the multiplicity cone, so their stationary equilibrium is unique. These cities experienced the same abrupt reduction in commuters as large cities did during the pandemic, but their dynamic evolution simply brings them back to the prepandemic stationary equilibrium. Hence, they exhibit no permanent change in the number of commuters. In sum, we argue that the pandemic acted as an equilibrium selection device for cities where multiple stationary equilibria existed *ex ante*. In a rare—in our view—empirical occurrence, the spatially uniform temporary push of most workers into a remote state during the pandemic has revealed the set of cities featuring multiple stationary equilibria and has served as a selection mechanism to determine the stationary equilibrium to which these cities are converging.

We contrast these predictions of our model with cell phone-based data on worker mobility from SafeGraph, available from before the pandemic until mid-2022. We first identify CBDs in all cities by finding the group of census tracts that receive the largest numbers of commuters prepandemic.² We then track the number of trips to these CBDs over time. The results are revealing. In New York, NY, for example, relative to prepandemic levels, the pandemic reduced the number of trips to the CBD by more than 80 percent. After the pandemic, the number of trips increased slightly, but it has now stabilized at about 40 percent of the prepandemic level. San Francisco, CA, shows a similar pattern. In contrast, Madison, WI, experienced a drop in the pandemic as large as New York but has recovered fully to prepandemic levels. The implications of the apparent multiplicity of stationary equilibria are present more broadly in the data. The largest US cities, with employment above 1.5 million workers, experienced a drop of 80 percent in trips to the CBD during the pandemic but now have stabilized at only 60 percent of prepandemic levels. In contrast, the smallest CBSAs, with employment below 150,000 workers, experienced a comparable drop in CBD trips during the pandemic but have now gone, on average, fully back to prepandemic levels.

The available housing data are also consistent with the mechanism in our theory. As others have documented, the pandemic generated a flattening of house price distance gradients in cities (Gupta, Mittal, Peeters, and Van Nieuwerburgh 2022). Namely, prices at the CBD declined relative to prices further away from the city. This is a natural consequence of the expansion of work from home since the advantage of living close to work is diminished if workers do not commute to the office or do so only a couple of days a week. What has not been documented but we document here, using Zillow zip code-level data, is that the flattening in the rent gradient has continued in the largest cities but has reversed and disappeared in the smaller ones. The cities where trips to the CBD have gone back to prepandemic levels, like Madison, or Durham, NC, exhibit housing price gradients that are not statistically different from prepandemic ones. In contrast, the larger cities where CBD trips have stabilized at much lower levels, like New York and San Francisco, exhibit a large and significant flattening of this gradient, which has been exacerbated since the pandemic.

²In some cases, we identify multiple CBDs when several areas have more than 80 percent of the commuters to the main CBD. One example is Midtown and Downtown in Manhattan.

What are the welfare implications of our findings? For small cities outside the multiplicity cone, the long-run welfare effect is zero. These cities have simply reverted to the prepandemic equilibrium. The transition lasted a couple of years, but the transition costs are probably only marginal. In contrast, for cities that have permanently switched to the work-from-home intensive stationary equilibrium, the welfare implications depend on the welfare ranking of both stable stationary equilibria. A direct comparison of these stationary equilibria in the 208 cities that we estimate are inside the multiplicity cone reveals that the welfare costs are positive but relatively modest. The costs are about 3.7 percent for Los Angeles and San Jose, 3.2 percent for New York, 2.8 percent for San Francisco, and 2 percent for Phoenix. A stationary equilibrium with office work is indeed better than the remote work intensive stationary equilibrium, but the benefits from the lack of commuting compensate for most of the losses. Of course, these numbers depend on the details of the estimation strategy of our model and the many sources of data, including the ACS and the NLSY, that we use to implement it. We discuss the details of the estimation of our dynamic discrete choice model in the body of the paper.

Our paper is related to several strands of the literature. First, we contribute to the emerging literature studying the effects of remote work on urban environments. Liu and Su (2023) document a decrease in the urban wage premium of occupations with higher levels of remote work since the COVID-19 pandemic, consistent with a decline in the agglomeration externality for these occupations. Davis, Ghent, and Gregory (2024) develop a model where remote work productivity increases via adoption externalities and study the response of remote work employment to improvements in its technology. Parkhomenko and Delventhal (2024) develop a quantitative spatial equilibrium model with work from home and commuting that predicts the optimal frequency of remote work. They study the spatial redistribution of economic activity induced by an increase in the amenity value of remote work. We propose an alternative interpretation to the large and persistent increase in remote work that does not rely on exogenous productivity or amenity shocks. Duranton and Handbury (2023) show that exogenous changes in which workers are allowed to work remotely in a monocentric city model can replicate observed changes in commuting and demand for housing. Instead, we show how the coordination of agents in a city can lead to multiple stationary equilibria and how temporary shocks to the number of commuters can permanently affect the structure of cities by selecting a different stationary equilibrium.

Remote work has been linked to changes in residential and commercial housing prices within and across cities (Mondragon and Wieland 2022; Gupta, Mittal, and Van Nieuwerburgh 2022; Brueckner, Kahn, and Lin 2021; Liu and Su 2021; Ramani and Bloom 2021; Althoff, Eckert, Ganapati, and Walsh 2021). Our finding that residential housing rent gradients with respect to distance to the CBD are diverging between larger and smaller cities is, to our knowledge, a new fact. We also provide an interpretation for this fact since these housing rent gradients map directly into structural parameters of our model.³ We show how the divergence between larger

³Duranton and Puga (2023) also link housing rent gradients to the elasticity of urban costs with respect to city size.

and smaller cities is predicted by our model, given that larger cities are more likely than smaller cities to converge to the remote work-intensive stationary equilibrium.

Understanding the productivity of remote work relative to in-person work remains an active area of research. Building on earlier studies (Oettinger 2011; Cappelli and Keller 2013), recent work indicates that remote workers may perform better than in-person workers in some contexts (Choudhury, Khanna, Makridis, and Schirmann 2022; Bloom, Han, and Liang 2022; Barrero, Bloom, and Davis 2021; Etheridge, Wang, and Tang 2020; Bloom, Liang, Roberts, and Ying 2015). However, workers employed in occupations involving a significant amount of problem solving tend to experience lower performance when working remotely (Emanuel, Harrington, and Pallais 2022; Kunn, Seel, and Zegners 2020). Gibbs, Mengel, and Siemroth (2023) study employee productivity before and during the COVID-19 pandemic among skilled professionals at an Indian information technology services company. They find that employee productivity fell by 8 percent to 19 percent and point to higher communication costs and longer time spent on coordination activities and meetings as sources for this decline. They also find that employees networked with fewer individuals and business units, both inside and outside the firm.⁴ Other studies emphasize that the higher earnings of remote workers can be partly explained by their selection into remote work based on unobservable characteristics (Harrington and Emanuel 2022; Atkin, Schoar, and Shinde 2022). Our model recognizes workers' heterogeneous preferences for remote work (Mas and Pallais 2017; Lewandowski, Lipowska, and Smoter 2026). Our analysis of the NLSY79 provides new estimates of the remote work premium from nonexperimental data that account for unobserved persistent heterogeneity in worker performance. Our findings indicate that the technology of remote work has been improving steadily over the past decades while remaining less productive than in-person work on average across occupations.

Agglomeration economies are a central tenet of urban economics (Henderson 2003; Combes et al. 2012). When cities successfully bring people together, positive spillovers abound. People interact, learn from each other, and foster entrepreneurship, leading to greater productivity within the city (Moretti 2004; Gennaioli, La Porta, Lopez-de Silanes, and Shleifer 2013; Glaeser and Maré 2001; Baum-Snow and Pavan 2012; De La Roca and Puga 2017).⁵ We contribute to the literature examining urban agglomeration economies by studying how remote work alters their workings. Canonical theories of cities predict that more central locations command higher house prices because of better accessibility to the locus of production (Alonso 1964; Muth 1969). By removing the need to travel across locations to reach the workplace, remote work deprives cities of one of their most fundamental organizational principles. The main insight from our analysis is that cities with strong agglomeration externalities, typically viewed as successful, are also the most at risk of dramatic shifts in their structure when people choose to stay home. Similar to us, Owens, Rossi-Hansberg, and Sarte (2020) uses coordination problems generated by

⁴See also Yang et al. (2022). They find that firm-wide remote work resulted in a less collaborative network among workers, with fewer connections between different parts.

⁵The productivity premium in larger cities has also been linked to the combination of more efficient firms and workers selecting these areas (Combes, Duranton, and Gobillon 2008; D'Costa and Overman 2014; Gaubert 2018).

residential externalities to rationalize the existence of depopulated and successful neighborhoods with similar underlying characteristics in Detroit.

The remainder of the paper is organized as follows. Section I presents stylized facts on remote work that emphasize the sudden increase in remote work since 2020, despite no large increase in the earnings premium for remote work. Section II documents the widely heterogeneous postpandemic experience across cities in trips to the CBD (Section IIA) and housing value gradients (Section IIB), before considering alternative explanations and presenting an intuitive case for multiple stationary equilibria (Section IIC). Section III presents our dynamic monocentric model with a choice of labor delivery mode. Section IV quantifies the model, including our estimation of the relative productivity of remote work and externality parameters. We evaluate the ability of our estimated model to rationalize the evolution of commuting across cities in Section V and compute the resulting welfare implications in Section VI. We relegate many of the model's and quantification's details, additional robustness exercises, and counterfactuals to the Supplemental Appendix.

I. The Evolution of Remote Work

We begin by documenting changes in the shares of workers delivering work remotely and changes in the hourly earnings' remote work premium between 1980 and 2023, accessed through IPUMS USA (Ruggles et al. 2025). We rely on two data sources. First, we use the decennial census waves of 1980, 1990, and 2000 and the annual American Community Survey waves between 2005 and 2023. From this source, we obtain large repeated cross-sectional information on the individual earnings per hour of workers who work in person or remotely. Second, we use the National Longitudinal Survey of Youth 1979 (NLSY79) from the US Bureau of Labor Statistics (2022), a panel of 4,147 individuals from which we collect information on the number of hours of remote work, hourly pay, and occupation every 2 years from 1998 to 2022. The repeated observations of the same workers allow us to control for permanent unobserved heterogeneity among workers and obtain estimates of the remote work premium that are not biased by selection on unobservable characteristics (Harrington and Emanuel 2022; Atkin, Schoar, and Shinde 2022). We provide details about the construction of our samples in Supplemental Appendix A.A2.

Remote work has become an increasingly common form of labor delivery over the past few decades.⁶ While part of this increase is likely due to improvements in communication technologies or remote work preferences, another part may arise via gradual changes in the composition of the labor force, both in terms of educational attainment and employment by occupation and industry. Our first exercise aims at removing the effect of compositional changes.

In Figure 1, panel A, we report time-varying estimates of a logit model for the probability of being a remote worker. We use census/ACS and NLSY data separately. In both datasets, we control for workers' education, gender, age, marital status, race, experience on the job, and broad industry and occupation fixed effects.

⁶Statistics typically report the share of all working hours performed at home over time (Oettinger 2011; Cappelli and Keller 2013; Liu and Su 2023).

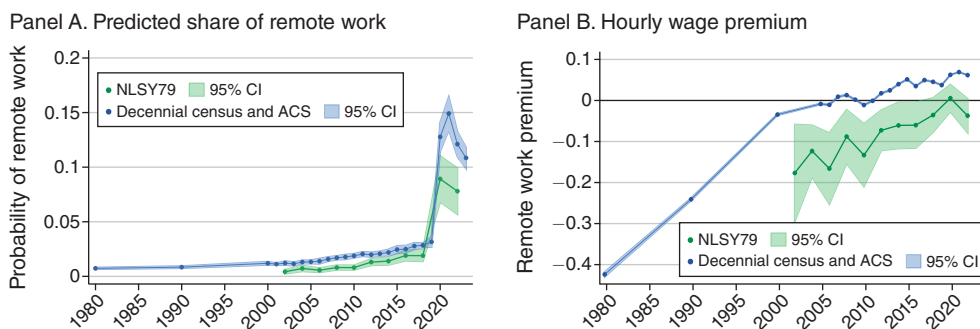


FIGURE 1. PREVALENCE AND EARNINGS PREMIUM OF REMOTE WORK

Notes: Panel A reports the predicted share of remote workers from a logit model with year fixed effects and controlling for workers' education, gender, age, marital status, race, experience on the job, and broad industry and occupation fixed effects. The estimates report the predicted probability that an individual with sample-average characteristics is a remote worker in any given year. Panel B reports the coefficients on yearly remote work indicators from a regression of log hourly wages on all the panel A controls plus year-specific effects for occupation, industry, education, region, and demographics. In the census/ACS data, "remote workers" are defined as those reporting working from home when asked about their means of transportation to work. In the NLSY data, "remote workers" are defined as those who work for more than 24 hours at home per week. Each point represents an estimate for a year. The shaded areas represent the 95 confidence interval around the estimates.

In the census/ACS data, we define as "remote workers" those who report working from home when asked about their means of transportation to work. In the NLSY data, we define as "remote workers" those who work for more than 24 hours (i.e., 3 days) at home in a week. We then predict the shares of remote workers, varying the year and setting the values of all covariates to their mean value over the sample; these shares can be interpreted as the probability of being a remote worker for someone with typical and time-invariant characteristics.⁷ Both datasets reveal a small and gradual rise in the share of remote work over time and a sudden postpandemic jump. In the ACS, this share increases from less than 1 percent in 1980 to 2.6 percent in 2019, jumping to 12 percent in 2020 and peaking at 15 percent in 2021, before returning to 11 percent in 2023. The NLSY features a similar pattern, increasing from 1.4 percent in 1998 to 3.7 percent in 2018, then rising to 9.2 percent in 2020, before dropping back to 7.8 percent in 2022. Importantly, since these patterns control for individual characteristics and compositional changes, the gradual increase is likely due to changes in technology or remote work preferences, but the large change in 2020 is probably related to the pandemic.

In Figure 1, panel B, we estimate the hourly wage premium of remote work relative to in-person work over time. In the census/ACS data, we regress log hourly wages on year-specific indicators of remote work, the same controls as those used in the logit model described above, and allowing in addition for year-specific effects for occupation, industry, education, region, and demographics. In the NLSY data, we complement the specification by adding individual fixed effects. Both exercises indicate a steady increase in the premium for remote work. Estimates from the ACS data indicate that, in 1980, remote work implied a discount of 44.5 percent in hourly

⁷In Table A1 of Supplemental Appendix A.A2, we report the actual share of remote workers over time as measured in the ACS.

rates compared to in-person work; in 2022, remote workers with similar observable characteristics earned 6.5 percent higher earnings relative to observationally equivalent office workers.⁸ In the NLSY sample, the discount for remote work went from 18.5 percent in 2000 to 3.1 percent in 2022. The stronger wage discount in the NLSY is consistent with remote workers having unobserved characteristics that make them higher earners, everything else equal. Remarkably, the remote work premium did not increase substantially between 2018 and 2022.⁹ This is consistent with the large increase in remote work being driven by other mechanisms rather than by a large productivity increase of remote work. The sudden increase in the probability of remote work, coupled with a stable wage discount, is consistent with the rapid shift to remote work being driven by mechanisms other than large remote work productivity or preference shocks. In the next section, we present evidence and articulate the case of multiple equilibria as such a mechanism.

II. Heterogeneous Responses across Cities

Individual decisions to reduce social contact during the pandemic and the related stay-at-home policies had dramatic consequences on the level and evolution of commuting and the value of proximity to the CBD across cities. We provide evidence of this impact using individual cell phone mobility data and estimates of the gradient of housing prices relative to distance to the CBD. We argue that, taken together, the evidence we provide can be rationalized only with mechanisms that allow cities to respond in drastically different ways to a common lockdown shock. Multiplicity of stationary equilibria provides such a mechanism.

A. *The Evolution of Commuting Patterns across Cities*

To analyze individual mobility, we use anonymized data from SafeGraph, Inc. (2022), a data provider that tracks foot traffic to businesses relying on mobile phone GPS locations from third-party applications. We use the Neighborhood Patterns dataset, which aggregates the amount of foot traffic at the census block group level, to study the monthly evolution of trips to the CBD.¹⁰ We first make use of the SafeGraph data to define the location of CBDs in each CBSA. The Neighborhood Patterns dataset provides information on the number of monthly visits recorded in each census block group. In particular, these visits are broken down by their most likely purpose, based on their observed characteristics.¹¹ We identify the center point of the CBDs as the center of a disc with a radius of 2 kilometers (km) that receives

⁸The ACS series stops in 2022 since the 2023 ACS did not report industry codes at the time of analysis.

⁹In Figure C2 of Supplemental Appendix C.C3, we show that this stability is also present separately for occupations with high and low teleworkable indices, as defined by Dingel and Neiman (2020).

¹⁰Chapple et al. (2023) also use SafeGraph data to document patterns of downtown foot traffic in 62 North American cities. They combine SafeGraph data with another private data source to obtain estimates until 2023. Their results confirm that, while downtown visits have recovered to prepandemic levels in some smaller cities, they have stabilized between 40 and 60 percent of their prepandemic levels in most larger cities, with little change between July 2022 and April 2023.

¹¹Workplace visits are defined as those recorded Monday through Friday between 7:30 AM and 5:30 PM and that dwelled for at least six hours. While the workplace visit count is missing for many CBSAs and census block groups in each CBSA, the coverage is sufficient to compute the version of Figure 2 and Figure B1 for office trips only in Supplemental Appendix B.B2. We find similar patterns.

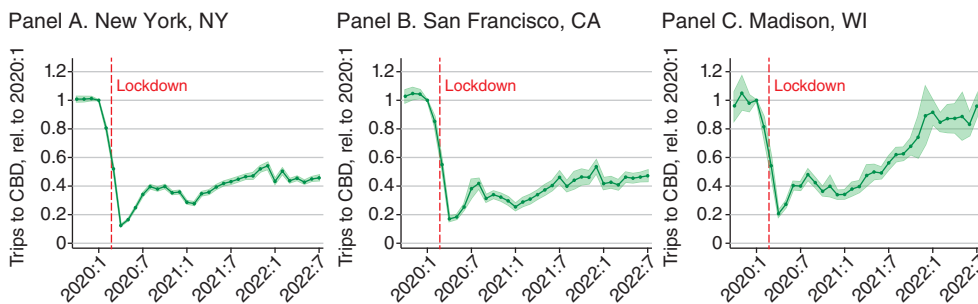


FIGURE 2. CHANGE IN VISITS TO THE CBD FOR SELECTED CITIES

Notes: Estimates of monthly indicators in a separate regression for each CBSA of visits to the CBD at the block group-month level. The outcome variable is the number of visits to the CBD from the specified block group, divided by the number of residing devices in that block group, as defined in Supplemental Appendix equation (A1), scaled by the value of this ratio in January 2020. The shaded areas represent the contour delimited by the consecutive 95 percent confidence intervals of each monthly indicator. Standard errors are clustered at the census tract of origin.

the largest amount of work-related foot traffic. We then include all the block groups within this 2 km disk into the CBD. We also allow cities to have several CBDs. We provide the details of our procedure in Supplemental Appendix A.A3.

The Neighborhood Patterns dataset also records the number of visits to each destination block group by block group of origin, allowing to compute bilateral flows of foot traffic at the monthly level.¹² On average, SafeGraph tracks around 30 million devices each month between January 2020 and July 2022.¹³ However, the number of mobile phones recorded in a city and the set of block groups of origin in which they are located vary across months. Some of the variation in the raw counts of foot traffic is therefore due to changes in the size of the sample and in the set of origin block groups included in the dataset. We control for these variations by analyzing the change in foot traffic flows at the origin block group level. In particular, we compute the share of monthly foot traffic originating from each block group that is directed to the CBD, we normalize the share of visits to the CBD from each origin block group by its value in January 2020 and then regress the normalized origin block group-level shares of visits to the CBD on month indicators.¹⁴

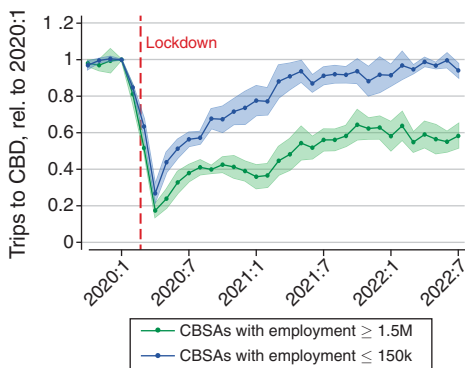
Figure 2 shows the volume of monthly trips to the CBD over the course of the pandemic, relative to its value in January 2020, for New York, San Francisco, and Madison. For all 3 cities, the first months of the pandemic saw a drastic drop in trips to around 20 percent of their initial value. The recovery, however, has been quite different. While Madison, a relatively smaller city, essentially went back to its prepandemic levels by the end of our sample period, CBD trips for larger cities like New York and San Francisco have stabilized at around 40 percent of their prepandemic levels, a dramatic drop.

¹²The block group of origin is imputed by assigning to each mobile phone the block group in which the phone is recorded at night as the block group of residence.

¹³In the largest cities (New York, Los Angeles, Chicago), between 400,000 and 800,000 devices are recorded each month.

¹⁴Chapple et al. (2023) measure foot traffic by counting visits to downtown businesses, scaled by the sampling rate of the state. Instead, we count visits from any block group to downtown block groups, scaled by the sampling rate of each origin block group. That is, we control for changes in sampling rates across neighborhoods of each city, which we view as more robust than using the state-level sampling rate.

Panel A. Large versus small CBSAs, monthly



Panel B. All CBSAs, 2020:4 versus 2022:7

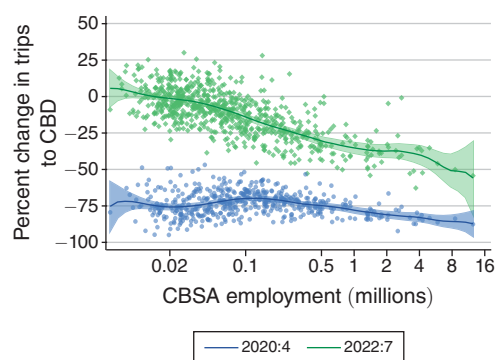


FIGURE 3. RESPONSE OF TRIPS TO THE CBD RELATIVE TO JANUARY 2020

Notes: Panel A reports the estimates of the average volume of visits to the CBD from block groups located in (i) CBSAs with total employment above 1.5 million (25 largest CBSAs) and (ii) CBSAs with total employment below 150,000 (663 smallest CBSAs) expressed as a share of their values in January 2020. The shaded areas represent the contour delimited by the consecutive 95 percent confidence intervals of each monthly indicator. Standard errors are clustered by month and census region. Panel B reports the change in the average level of visits to each CBSA's CBD between January 2020 and April 2020, and between January 2020 and July 2022, against employment (BEA). The solid lines represent a fitted kernel-weighted local polynomial smoothing, with the shaded area corresponding to the 95 percent confidence interval around the fitted value.

This behavior is, in fact, systematic. Panel A of Figure 3 shows the monthly trips to the CBD relative to January 2020 for cities grouped by size. Our measure of size is total nonfarm employment in the CBSA in 2019, published by the Bureau of Economic Analysis (2023).¹⁵ The green line reports the estimates of the trips to the CBD for the largest CBSAs with employment of more than 1.5 million, while the blue line collects the smallest CBSAs in our sample, with employment below 150,000.¹⁶ We normalize to 1 the number of trips in January 2020. Both groups experienced a sharp decline to less than 30 percent of prepandemic trips up to April 2020, but their paths are remarkably different afterward; while the group of small CBSAs returned to their prepandemic levels of visits to the CBD around early 2021, large CBSAs stabilized at around only 60 percent of their initial level.

Panel B of Figure 3 considers the relation of trips with city size in more detail. The first series shows the extent to which these trips declined in April 2020 compared to January 2020, plotted against the city's total employment. Trips to the CBD declined by about 75 percent of the prepandemic level on average, with only little variation in the experiences of different cities. Importantly, this initial drop does not appear related to city size. In contrast, city size is systematically related to the return to office for the latest available data. The second series shows that in July 2022,

¹⁵This measure has the advantage of being available for all CBSAs, allowing us to describe patterns of recovery for the largest set of CBSAs—847 in total. In Supplemental Appendix B.B6, we reproduce the key graphs from Section II using as size the measure of employment used in the quantification, computed from census microdata, and consistent with our measure of commuting. We confirm that the patterns described below hold for the subset of CBSAs for which we can measure employment in the census.

¹⁶The smallest CBSA with employment above 1.5 million is Austin, Texas, while the largest CBSA with employment below 150,000 is Rochester, Minnesota.

the largest cities are substantially more likely to have trip levels way below their prepandemic levels.

B. Distance Gradients in House Prices

The previous section showed that the evolution of flows to the CBD has been heterogeneous across cities. Of course, changes in remote work patterns imply changes in demand for housing within cities. As workers spend less time on-site, a short commute becomes less important, leading to lower demand for housing next to centers of economic activity, and, everything else equal, a lower premium for proximity to the CBD.

Changes in urban land gradients may be driven by a large number of factors varying at the CBSA level. Some of these include constraints on local housing development (Gyourko, Saiz, and Summers 2008; Saiz 2010; Lutz and Sand 2022) or the connectivity between urban and suburban parts of the CBSA.¹⁷ For this reason, we estimate the CBSA-specific price and rent gradients in our bid-rent regression. We use monthly zip code-level housing price data from Zillow (2025).¹⁸ These high-frequency data allow us to follow the monthly changes in housing prices for many cities since 2019.

Indexing CBSAs by j , zip codes by i , and months by t , we estimate the monthly slope of the bid-rent function, using the following empirical specification,

$$(1) \quad \ln p_{ijt} = \delta_{jt} \ln(\text{dist}_{ij}) + \beta X_{ij} + \alpha_{jt} + e_{ijt},$$

where units of observation are zip code-months. We refer to p_{ijt} as the price in zip code i of CBSA j in month t , and dist_{ij} is the distance in kilometers between the centroid z_{ij} of zip code i and the CBD of CBSA j . We include CBSA-month fixed effects, α_{jt} , and zip code-level control variables, X_{ij} . The zip code controls contain the log of median household income, the share of households in the top national income quintile, the median age, and the share of Black residents, all measured in 2019, using ACS ZIP Code Tabulation Area tables from the US Census Bureau (2020a) and geographic controls.¹⁹

Figure 4 plots the time evolution of $-(\delta_{jt} - \delta_{jt_0})$, the drop in the distance coefficient at time t relative to the baseline of January 2020; a decrease in the reported coefficients constitutes a flattening of the price gradient. Consistent with the evidence presented in Figure 2, we find that the willingness to pay for proximity to the CBD is permanently lower in New York and San Francisco, while it barely starts decreasing before returning to its initial values in Madison.

Panel A of Figure 5 shows the average distance elasticity of house prices relative to January 2020 for the same group of city sizes as in Figure 3. In particular, the vertical axis reports again $-(\delta_{jt} - \delta_{jt_0})$, the drop in the distance coefficient at time t relative to the baseline of January 2020, so that a decrease in the reported coefficients

¹⁷ See Bailey et al. (2018) for measures of social connections using Facebook data.

¹⁸ The monthly housing price dataset is the Zillow Home Value Index (ZHVI). Zillow also publishes the Zillow Observed Rent Index (ZORI), a monthly housing rent data that would be a closer analog to the ACS rent data. However, the ZORI is reported for fewer zip codes, making the estimation of the gradient imprecise for many CBSAs.

¹⁹ The geographic controls are the tract-level controls in footnote 47, averaged by zip code, using tract population as weights.

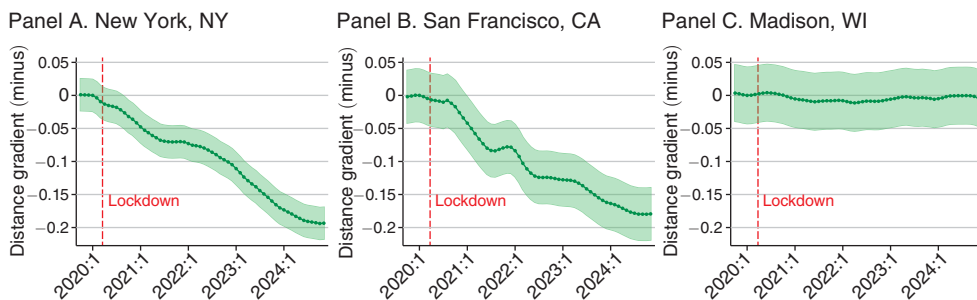


FIGURE 4. CHANGE IN HOUSING VALUES GRADIENTS FOR SELECTED CITIES

Notes: Estimates of monthly coefficients for log distance between zip codes and the CBD, relative to their value in January 2020. In a separate regression for each CBSA, we regress log housing values at the zip code level, on log distance by month, and geographic controls. The shaded areas represent the 95 percent confidence intervals of each monthly indicator, computed from robust standard errors. Data on housing values from Zillow's ZHVI index.

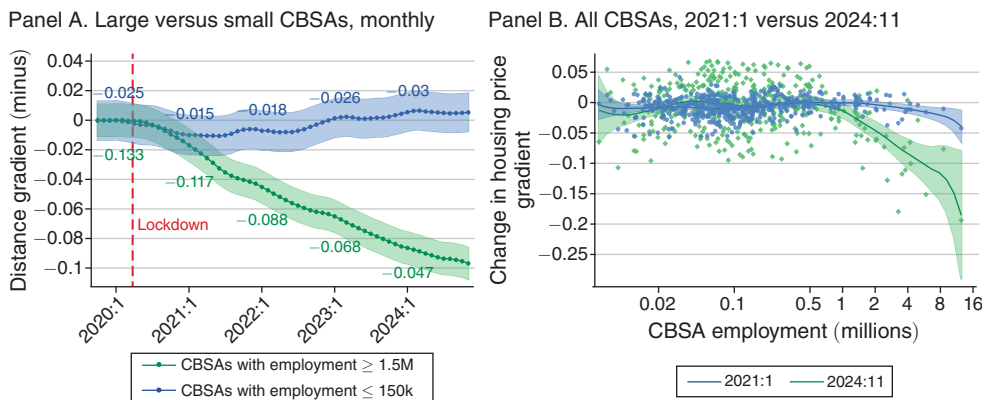


FIGURE 5. RESPONSE OF HOUSE PRICE GRADIENTS RELATIVE TO JANUARY 2020

Notes: Panel A reports the difference between the average housing price gradients in January 2020 and each following month, using the same employment thresholds as in Figure 3. In this figure, 25 cities have employment above 1.5 million, and 225 have employment below 150,000. A negative value for the difference in housing price gradients implies a lower premium for housing located close to the CBD. The value of the average housing price gradient in each group is reported next to the marker of the associated group every six months. The shaded areas represent the contour delimited by the consecutive 95 percent confidence intervals of each monthly indicator. Panel B reports the difference between the housing price gradients in January 2020 and January 2021, and November 2024, respectively, against employment (BEA). Each marker represents a CBSA. The gradients are estimated using zip code level data from Zillow for 278 CBSAs. The solid line represents a fitted kernel-weighted local polynomial smoothing, with the shaded area corresponding to the 95 percent confidence interval around the fitted value.

implies a flattening of the price gradient. Quite naturally, prices move more slowly than trips over time. All cities experience a flattening of the housing price gradients of around 0.01 by January 2021. This decline stops and reverses for small cities, which return to around their prepandemic level; the drop in the gradient of larger cities, instead, persists and appears to stabilize around 0.095 by the end of 2024.

Panel B of Figure 5 illustrates the cross-sectional variation as a function of city size in the drop of housing price gradients at two different points during and after

the pandemic. As above, the vertical axis represents the drop in the distance coefficient at time t relative to the baseline of January 2020. During the first year of the pandemic (January 2020 to January 2021), the distance elasticity of housing prices flattened by about 0.01 on average. This drop is sizable relative to the average price gradient in January, equal to -0.044 . The heterogeneity is moderate and again unrelated to city size. Consistent with evidence on CBD trips, however, the flattening in the price gradient becomes negatively associated with city size as the pandemic progresses and then recedes.

C. Potential Explanations

The dramatic changes in large cities documented in the last two subsections have received the most attention from researchers. However, our results stress that not all cities have followed the same path after the COVID-19 pandemic commuting shock. In particular, the return of smaller cities to their prepandemic structure is a salient feature of the postpandemic experience as well.

Existing explanations for the sharp increase in remote work rely on large and sudden increases in either aggregate remote work productivity or worker preferences (or a combination thereof).²⁰ These rapid and large technology and preference changes would be expected to produce at least some noticeable and lasting shifts in commuting patterns across many cities. Yet our evidence shows that a sizable group of smaller CBSAs have reverted to their prepandemic structures, suggesting that these national shocks, while probably also relevant, cannot fully account for the observed heterogeneity. Of course, cities differ in their share of employment amenable to remote work, so even a large nationwide improvement in remote work productivity or attractiveness might affect remote work adoption only moderately in cities with low shares of teleworkable employment. However, Dingel and Neiman (2020) estimate that this share varies from 25 to 55 percent across CBSAs. Hence, at least one-fourth of jobs remain teleworkable even in the cities least amenable to remote work, while in the city most apt to remote work, almost one in two workers are employed in non-teleworkable occupations. In our data, cities that fully returned to prepandemic CBD trips by July 2022, and for which there is little to no change in the willingness to pay for proximity to downtown by the end of 2024, still have 34 percent of workers employed in teleworkable occupations. While large country-level changes in preferences or productivity would normally be expected to imply some persistent adjustment in trips to the CBD or distance-rent gradients even in small cities, we find little evidence of such changes. Importantly, the share of teleworkable occupations in cities with a permanent decline of at least 40 percent in trips to CBD is 43 percent—a higher, but not dramatically higher, value to explain the large differences in commuting patterns across cities. Although plausible, explanations based solely on preference or technology shocks appear insufficient to fully account for the range of heterogeneous behaviors we document.

²⁰ Davis, Ghent, and Gregory (2024) calibrate a short-term increase in remote work productivity of 83 percent among high-skilled workers. Parkhomenko and Delventhal (2024) opt for a calibration combining a rise in remote work productivity of about 10 percent with a decline in the aversion to remote work of 46 to 70 percent.

A related series of candidate explanations suggest that cities differ in other characteristics, unrelated to occupational composition, which produce differential evolution of commuting patterns. One possibility, for example, is that large cities can be congested, making commuting to the city center costly and workers more willing to transition to remote work. If congestion were the main driver of these patterns, we might expect a lower propensity to travel to the CBD even before the pandemic. However, the data show the opposite pattern. In particular, over the last 3 months of 2019, 30.6 percent of all trips in CBSAs with employment above 1.5 million were directed toward the CBD, compared with only 15.6 percent in CBSAs with fewer than 150,000 workers. This higher rate of CBD trips in larger CBSAs is consistent with the idea that those cities' central areas are congested precisely because workers want to have access to the wages and benefits they provide.

A second possibility is that the differential behavior of large and small cities is driven by differential employment concentration across establishments. In particular, if smaller CBSAs are more likely to host large establishments, they may be more likely to return to high levels of commuting if those establishments participate in bringing workers back to the office. We examine this possibility in Supplemental Appendix B.B4 by estimating for each CBSA the share of employment accounted for by the largest 5 percent and 10 percent of establishments. We find that employment is in fact *less* concentrated in smaller cities than in larger ones. We interpret this finding as suggesting that differences in the return to the office between large and small CBSAs are not primarily driven by the idiosyncratic decisions of the largest establishments.

Another possibility is that, during the pandemic, workers shifted toward occupations that are more amenable to remote work. This shift might be driven, for example, by national changes in preferences or technology. However, we see no footprint of these changes in the wage premium by occupations. In Supplemental Appendix C.C3, we reestimate the wage premium dividing occupations according to how amenable they are to remote work, using the Dingel and Neiman (2020) index. Supplemental Appendix Figure C2 shows no differential evolution of this premium over the pandemic years. Regardless of the evolution of the wage premium, we could still observe the patterns described in the sections above if employment shifted toward teleworkable occupations at a higher rate in larger than in smaller cities. In Supplemental Appendix B.B5 we examine this possibility in detail. We find that, in the wake of the pandemic, the share of teleworkable employment increased by about 5 percent, a value much smaller than the large drops in foot traffic; crucially, by the end of 2023, the share of teleworkable employment increased *more* in smaller CBSAs (7.9 percent) than in larger ones (5.8 percent). This finding makes it unlikely that differential changes in the occupational composition can explain the patterns we have described.

While all plausible, these arguments seem limited in their ability to jointly reconcile the range of evidence we present. This evidence highlights the need for a theory that rationalizes heterogeneous responses across cities to a common shock to the number of commuters. The theory we propose rationalizes heterogeneous responses depending on whether a city exhibits multiplicity of stationary equilibria, which in turn depends on the balance between agglomeration externalities, occupational composition, and commuting costs. In particular, when externalities are the main

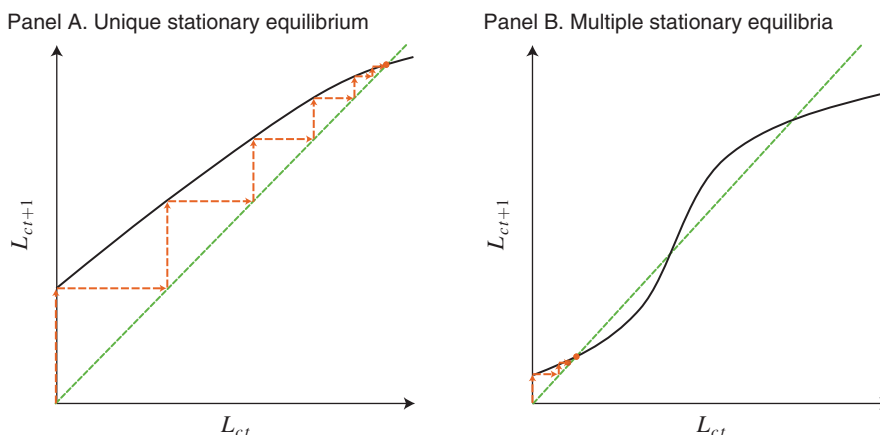


FIGURE 6. HETEROGENEOUS RESPONSES TO LOCKDOWN SHOCKS

Notes: This figure shows the adjustment toward the stationary equilibrium in the case of unique versus multiple stationary equilibria. In both panels, the orange line depicts the adjustment toward a stationary equilibrium following an initial shock that brings the number of commuters $L_{c,t}$ to zero. The dashed gray line in the right panel shows the 45-degree line. When the stationary equilibrium is unique (left panel), the city returns to the initial stationary equilibrium. With multiple stationary equilibria, the city converges to the low-commuting one.

reason for downtown's attractiveness and so a city features multiplicity, high levels of prepandemic commuting were only one possible stationary equilibrium. In those cases, the pandemic shock can move the city to a dynamic path that converges to a stationary equilibrium with much higher levels of remote work.

To illustrate this intuition, Figure 6 presents a phase diagram for the dynamics of the number of commuters in our model. The axes represent the number of commuters in two adjacent periods. All stationary equilibria are represented as crossing with the 45-degree line, where the number of commuters (the state variable) is constant over time. We compare how a city with a single stationary equilibrium (left panel) and a city with multiple stationary equilibria, initially in the high-commuting stationary equilibrium (right panel), respond to the same commuting shock.²¹

From the stationary equilibrium with high levels of commuting, this shock initially reduces the measure of commuters $L_{c,t}$ to almost zero. Subsequently, as the restrictions are lifted and public confidence returns, individuals reoptimize their labor delivery mode. The set of incentives workers face, however, is now different between the two city structures.

In cities with a unique equilibrium, the aggregate number of commuters $L_{c,t}$ has limited influence on the net individual benefit of working in the office. In this environment, technology is the primary determinant of wage differentials, and the amenity value of being downtown is largely independent of the CBD's overall vibrancy. Because externalities have little impact on individual choices, the city's fundamentals

²¹ Here, and in all of the phase diagrams below, we implicitly assume a unique dynamic equilibrium. Although we cannot exclude the presence of multiple dynamic equilibria, Supplemental Appendix E.E6 provides extensive testing suggesting that this possibility is limited in practice.

ultimately restore the initial equilibrium, as depicted by the path of the orange line in panel A.

In contrast, the economy's fundamentals are not the sole determinant of the final aggregate state of cities with multiple stationary equilibria. In this case, the current number of commuters powerfully shapes the incentives to return to the office. A lockdown pushing the city into low commuting levels significantly alters the individual productivity differential in favor of remote work. While travel congestion may decrease, the allure of the downtown area also diminishes, and the amenity value of working in the central business district vanishes. As illustrated by the orange line in panel B of Figure 6, the city finally settles into a low-commuting stationary equilibrium. Since the number of commuters is permanently lower, the rent gradient is also flatter and closer to zero. This logic explains how the same temporary shock can have large permanent effects in some cities and no effect in others. It also explains why prepandemic changes in preferences and technology that did not change the stationary equilibrium only led to marginal effects on the number of commuters.

Equipped with this intuition, we set up the formal model in Section III and derive testable predictions on the existence of multiple equilibria for multiple cities. We then investigate in Section IV whether cities like Madison are predicted to respond like panel A of Figure 6, whether cities like New York and San Francisco are predicted to respond like panel B, and the welfare and cross-city implications of these different responses.

III. Model

A. Overview

We model a disk-shaped city with an exogenous population of size L .²² Time is discrete. A worker starts period t in a labor delivery mode $\ell \in \{c, m\}$ (commuting or remote) that was determined in the previous period and is fixed within the current one. The worker chooses a residence location, housing quality, and consumption, supplying labor according to mode ℓ . Workers then observe idiosyncratic shocks $\varepsilon_t = \{\varepsilon_{ct}, \varepsilon_{mt}\}$ for each mode, make a labor delivery decision for the next period, and enjoy the corresponding idiosyncratic benefit. These benefits are drawn from a Gumbel distribution with location parameter $\tilde{\gamma}$ (the Euler constant) and scale parameter s , $G(\varepsilon) = \exp\{-e^{\varepsilon/s - \tilde{\gamma}}\}$. The period then ends, and in the next one, the new composition of labor delivery mode becomes predetermined.

The city is radially symmetric around a CBD, located at $d = 0$. Each unit of land (or housing plot) accommodates one resident, so the population density equals land density and integrates to the total population L . All plots located at the same distance d from the CBD are identical; we therefore index locations by their distance $d \in (0, \bar{D}]$ from the center, where $\bar{D}$ is the outer boundary of the city.

Production occurs in the CBD. Land is owned by absentee landlords, who can convert it into housing or use it for an alternative activity yielding an exogenous

²²In Supplemental Appendix B.B3 we explore possible systematic relationships between population changes and trips to the CBD. We find that population changes have been too limited, and too uncorrelated with post-COVID-19 behavior, to challenge the plausibility of this assumption.

return r_a . A worker residing on a housing plot at location d pays a rent $r(d)$ per unit of housing quality. The landlord supplies the chosen quality level by exerting effort proportional to the total rent minus the value of land. Hence, their net return per plot is $r(d)$, regardless of the tenant's choice of quality.²³

In what follows, we describe the production environment and then formalize the worker's problem. The within-period part determines indirect flow utilities and the city rent schedule, while the dynamic part governs the transitions across labor delivery modes and, consequently, the aggregate evolution of the city. We conclude with a characterization of the city's equilibrium.

B. Production

There are two goods in the economy: final consumption and housing. Final consumption is a homogeneous good produced in the CBD by profit-maximizing firms under constant returns to scale: One efficiency unit of labor produces one unit of good, at a constant marginal cost of $\bar{w}$.²⁴ The number of efficiency units of a worker, and hence her total wage w_ℓ , both depend on the chosen labor delivery mode ℓ . In particular, define

$$(2) \quad \tilde{L}_c \equiv L_c^\mu L^{1-\mu}$$

as the aggregate measure of labor units delivered in the city center, a function of the current number of commuters L_c , the total city size L , and the fraction of days per week a remote worker spends at home, μ . Then, the wage of commuters and remote workers (really *mixed* workers since they spend only a fraction of their work time at home) is given by²⁵

$$(3) \quad w_\ell \equiv \begin{cases} w_c \equiv \bar{w} \cdot A \tilde{L}_c^\delta & \text{if } \ell = c, \\ w_m \equiv \bar{w} \cdot z^\mu (A \tilde{L}_c^\delta)^{1-\mu} & \text{if } \ell = m. \end{cases}$$

In particular, the efficiency units offered by a commuting worker depend on the in-office productivity, $A > 0$, and on how many workers are on average in the CBD, $\tilde{L}_c$, with an elasticity $\delta > 0$. The efficiency units of a remote worker are a combination of the efficiency of remote work $z > 0$ and the efficiency of commuting work, weighted by the fraction of time remote workers spend on each mode. Free entry in production implies $\bar{p} = \bar{w} = 1$, after a normalization.

²³ Given that landlords are absent, an isomorphic interpretation is that the tenant pays rent in goods and the landlord uses some of these goods, rather than effort, to supply the demanded housing quality. Equivalently, the household could pay the landlord for the land and use the rest of the housing expenditure to improve housing quality directly.

²⁴ Our model abstracts from heterogeneous occupations. This choice is motivated by the empirical findings, in Supplemental Appendix B.B5, that transitions between occupations differentially amenable to remote work are too rare to be an empirically relevant margin of adjustment at our time horizon.

²⁵ In Supplemental Appendix C.C5, we show that this wage structure is the outcome of a linear production function in remote and in-person workers. More generally, if we use a CES production function and estimate the elasticity of substitution between these two types of labor, we find very large values. This finding motivates our linear production function.

Landlords produce housing quality on the plots they own. To supply H units of quality on a plot at location d , a landlord must expend $r(d)(H - 1)$ efficiency units of work, which has value $r(d)(H - 1)$ in the market since wages per efficiency unit are normalized to 1.²⁶ The resident pays $r(d)H$ for the unit, so the landlord's net return from the plot is $r(d)$, independent of the chosen quality level.²⁷

C. Problem of the Worker

A worker starts period t in state $\ell \in \{c, m\}$, solves a within-period problem delivering a flow utility u_ℓ , observes idiosyncratic shocks associated with the next period's labor delivery mode options, and makes a new choice ℓ' .

Within-Period Problem.—We write the within-period problem of a worker in state ℓ , in a city with L_c commuters, as

$$(4) \quad u_\ell(L_c) = \max_{Q_\ell, H_\ell, d} \log \left(T^{\mu_\ell} \left(\frac{B(\tilde{L}_c)}{\tau(\tilde{L}_c) d^\gamma} \right)^{1-\mu_\ell} Q_\ell^\alpha H_\ell^{1-\alpha} \right),$$

st

$$(5) \quad \mu_\ell \equiv \begin{cases} 0 & \text{if } \ell = c \\ \mu & \text{if } \ell = m \end{cases}$$

$$(6) \quad w_\ell = Q_\ell + r(d)H_\ell,$$

where we have omitted the time index and consider distances $d > 0$.

The parameter $\mu \in (0, 1]$ denotes the fixed fraction of workdays spent at home by remote workers. Commuters spend no time at home ($\mu_c = 0$). Agents take μ as a fixed job characteristic that they take as given when choosing to work remotely.

The worker chooses consumption Q_ℓ , housing H_ℓ , and location d . Amenity and disutility terms include $T > 0$, a benefit of working from home; $B(\tilde{L}_c)$, a benefit of visiting the CBD that increases with $\tilde{L}_c$; $\tau(\tilde{L}_c)$, the congestion cost of commuting, also increasing in $\tilde{L}_c$; and commuting costs d^γ , where $\gamma > 0$ measures the elasticity of commuting costs with respect to distance. The exponent $\alpha \in (0, 1)$ is the weight of consumption in utility. Workers earn a wage w_ℓ described in (3) and spend it on consumption, with a price normalized to 1, and housing, with unit price $r(d)$.

Our formulation captures the key trade-offs between commuting and remote work. On the amenity side, remote work allows an individual to enjoy the flexibility of working from home and paying lower commuting and congestion costs, at the price of lower enjoyment of the social and consumption opportunities in the CBD, all for a fraction μ of the time. On the wage side, remote work trades off the positive externalities arising from in-office interactions with the efficiency of communication technology z , again for a fraction μ of the time. The scale of economic activity in the CBD, as measured by $\tilde{L}_c$, shapes the balance of these trade-offs: When more

²⁶If $H < 1$, the landlord is effectively compensated for supplying less quality, for instance, through greater leisure.

²⁷This formulation of the housing sector—with unit-size plots and adjustable quality—allows us to combine a Cobb–Douglas within-period utility function with a tractable rent schedule based solely on rent per unit arbitrage. These two features make it possible to derive a clear analytical characterization of equilibrium in Section IIID.

agents visit the city center, individual productive efficiency is higher, the CBD is more lively, but congestion is heavier.

Dynamic Problem.—The solution to the within-period problem described above delivers the within-period utility flow. The worker then observes two new idiosyncratic shocks for the next period's labor delivery mode, $\varepsilon_{\ell'}$, and chooses ℓ' accounting for the evolution of the state of the economy. In particular, the dynamic decision of a worker who starts the period in state ℓ is

$$(7) \quad U_{\ell}(\{\varepsilon_{ct}, \varepsilon_{mt}\}; L_{ct}, \omega^t) = u_{\ell}(L_{ct}) + \max_{\ell' \in \{c, m\}} \left\{ \beta V_{\ell'}(L_{ct+1}; \omega^{t+1}) - F_{\ell\ell'} + \varepsilon_{\ell'} \right\},$$

where $\omega^t \equiv \{\omega_k\}_{k=t}^{+\infty}$ and $\omega_t \equiv \{A_t, z_t, F_{\ell\ell'}, T_t, B_t(\cdot), \tau_t(\cdot)\}$ is a known exogenous sequence of (possibly) time-varying technology, fixed costs, and amenities and

$$(8) \quad V_{\ell}(L_{ct}; \omega^t) = \int_{\varepsilon_m} \int_{\varepsilon_c} U_{\ell}(\{\varepsilon_c, \varepsilon_m\}; L_{ct}, \omega^t) dG(\varepsilon_c) dG(\varepsilon_m)$$

is the expected utility of a worker starting in state ℓ if L_{ct} workers are commuting. In these expressions, $F_{\ell\ell'}$ is the cost of transitioning from mode ℓ to mode ℓ' at time t .²⁸ Although the theory can accommodate any nonnegative values, we will set $F_{\ell\ell} = 0$ for $\ell \in \{c, m\}$. The term ε_{ℓ} is the idiosyncratic shock received by the worker in mode ℓ .

D. Equilibrium

We start by defining a within-period equilibrium condition that has to hold at each time, given the current distribution of commuters and remote workers. We then define a dynamic equilibrium, which describes the evolution of the economy when the within-period equilibrium holds at each time, and forward-looking workers make commuting and remote work choices based on—and consistent with—the aggregate evolution of the economy. Finally, we define a stationary equilibrium as a dynamic equilibrium where all endogenous variables are constant.

Within-Period Equilibrium.—We now define a within-period equilibrium. In the section, we eliminate the time index to simplify notation.

DEFINITION 1: *Given L_c and ω , a within-period equilibrium is a rent function $r(d)$, $d \in (0, \bar{D}]$, and an allocation $\{Q_c, Q_m, H_c, H_m\}$ such that (i) individuals solve (4)–(6), (ii) firms maximize profits and free entry drives profits to zero, and (iii) the housing market clears.*

²⁸ Note that, in the flow utility, the willingness to pay to live close to the CBD is a function of the labor delivery mode choice; agents will enjoy the remote work utility only if they pay the associated switching cost. As in standard urban models, the average utility of agents is equalized *across* locations where commuters live, and *across* locations where remote workers live, as a consequence of the first-order condition for residence location choice, d ; the average flow utility, however, is not equalized between commuters and remote workers. Hence, the fixed cost F should be interpreted as jointly capturing the cost of changing labor delivery mode and of switching the residence location between any location where commuters live and any location where remote workers live. Ultimately, it represents the cost of accessing the average flow utility of a different state.

We construct a within-period equilibrium by first solving for consumption and housing demand conditional on a distance and then letting the agent pick the distance that delivers the highest (indirect) utility. From the Cobb-Douglas structure, we have

$$(9) \quad Q_\ell = \alpha w_\ell,$$

$$(10) \quad H_\ell(d) = \frac{(1 - \alpha) w_\ell}{r(d)},$$

which implies an indirect utility of

$$(11) \quad u_\ell = \log \left(\bar{\alpha} T^{\mu_\ell} \left(\frac{B(\tilde{L}_c)}{\tau(\tilde{L}_c) d^\gamma} \right)^{1-\mu_\ell} \frac{w_\ell}{r(d)^{1-\alpha}} \right),$$

with $\bar{\alpha} \equiv \alpha^\alpha (1 - \alpha)^{1-\alpha}$, for $\ell \in \{c, m\}$. Taking the first-order conditions for the optimal distance to the CBD yields

$$(12) \quad \frac{r'_\ell(d)d}{r_\ell(d)} = -\frac{\gamma(1 - \mu_\ell)}{(1 - \alpha)} \implies r_\ell(d) = k_\ell d^{\frac{\gamma(1-\mu_\ell)}{(1-\alpha)}},$$

where $r'_\ell(d)$ is the slope of the rent function in the segment relevant to workers with delivery mode ℓ and k_ℓ is a constant of integration. At the optimal residence location, the savings on rents of a marginally farther location exactly compensate for the increased disutility in travel time. Since remote workers suffer less for a given travel time increase (by equation (5), $\mu_m \equiv \mu \in (0, 1] > \mu_c = 0$), their willingness to pay for proximity to the CBD is lower. Hence, commuters live in $(0, \bar{d}]$, remote workers live in $[\bar{d}, \bar{D}]$, and the equilibrium rent function has a steeper (constant) elasticity for commuters than for remote workers.

In our formulation, these conditions translate into a constant elasticity rent, with constants of integration k_c and k_m . These constants of integration k_c and k_m are characterized by two no-arbitrage conditions. First, the rent per unit of housing in the last residence location has to coincide with the alternative land use r_a , so that $r_m(\bar{D}) = r_a$. Second, land owners must be indifferent between renting to commuters and remote workers at $d = \bar{d}$, so that $r_c(\bar{d}) = r_m(\bar{d})$. Using these two conditions, the equilibrium rent function is

$$(13) \quad r(d) = \begin{cases} (\bar{D}/d)^{\gamma/(1-\alpha)} (\bar{d}/\bar{D})^{\mu\gamma/(1-\alpha)} r_a & \text{if } 0 < d < \bar{d} \\ (\bar{D}/d)^{(1-\mu)\gamma/(1-\alpha)} r_a & \text{if } \bar{d} \leq d \leq \bar{D} \end{cases}.$$

An implication of this result is that the average elasticity of rent to distance κ is given by

$$(14) \quad \kappa \equiv \frac{1}{\bar{D}} \int_{x=0}^{\bar{D}} \frac{\partial r(x)}{\partial x} \frac{x}{r(x)} dx = -\frac{\gamma}{1-\alpha} \left[(1-\mu) + \mu \frac{\bar{d}}{\bar{D}} \right].$$

The housing market clears at each distance because landowners rent one unit of land per resident at a rate $r(d)$ per unit of housing *quality*.²⁹ Each household occupies one plot of land but chooses its optimal housing quality $H_\ell(d)$ according to (10), so that total rent payments per household equal $r(d)H_\ell(d)$. Because the number of plots is fixed, the total measure of occupied plots in any region of the city must equal the number of residents living there. Hence, in equilibrium, the land density equals the population density, and the total number of plots within radius $\bar{d}$ must match the number of commuters L_c occupying that area.

In this circular city, the total number of units in the disk of radius $\bar{d}$ is $\pi\bar{d}^2$ and must equal the number of residents L_c in that disk. Similarly, $\pi\bar{D}^2 = L$. These conditions imply that $\bar{d} = (L_c/\pi)^{1/2}$ and $\bar{D} = (L/\pi)^{1/2}$. Hence, the rent gradient reflects the strength of distance frictions, as measured by γ , and the average value of being close to the CBD, as proxied by the commuting population L_c .

A within-period equilibrium is then described by this rent function, and the allocation implied by equations (9)–(10). The indirect utility function (11) then becomes

$$(15) \quad u_\ell = \begin{cases} \log \left[r_a^{\alpha-1} \bar{\alpha} \left(\frac{B(\tilde{L}_c)}{\tau(\tilde{L}_c)} \right) w_c / \left(\bar{d}^\mu \bar{D}^{1-\mu} \right)^\gamma \right] & \text{if } \ell = c \\ \log \left[r_a^{\alpha-1} \bar{\alpha} T^\mu \left(\frac{B(\tilde{L}_c)}{\tau(\tilde{L}_c)} \right)^{1-\mu} w_m / \bar{D}^{\gamma(1-\mu)} \right] & \text{if } \ell = m \end{cases}.$$

These indirect utilities of each labor delivery mode are an element of the dynamic problem of a worker, to which we now turn.

Dynamic Equilibrium.—A worker starts the period in state ℓ and enjoys the utility flow u_ℓ . The worker then observes two new idiosyncratic shocks $\varepsilon_{\ell t}$ and decides the new state for next period ℓ' , accounting for the exogenous evolution ω^t and the aggregate measure of commuting workers, L_{ct+1} .

The evolution of the measure of commuting workers is governed by the law of motion³⁰

$$(16) \quad L_{ct+1} = L_{ct}\lambda_{cct} + (L - L_{ct})(1 - \lambda_{mmt}),$$

where the share of workers in state ℓ transitioning to ℓ' is given by

$$(17) \quad \lambda_{\ell\ell'} = \frac{\exp(\bar{\varepsilon}_{\ell\ell'}(L_{ct+1})/s)}{\exp(\bar{\varepsilon}_{\ell ct}(L_{ct+1})/s) + \exp(\bar{\varepsilon}_{\ell mt}(L_{ct+1})/s)}.$$

²⁹From (4) and (13), for commuters ($\mu_c = 0$), we have $u_c(d) \propto \log[d^{-\gamma}r(d)^{-(1-\alpha)}] = \log(d^{-\gamma}d^\gamma) = \log(1)$, so $\lim_{d \rightarrow 0} u_c(d)$ is finite, and utility is continuous at the center. For remote workers ($\mu > 0$), $u_m(d) \propto \log[d^{-\gamma(1-\mu)}r(d)^{-(1-\alpha)}] = \log(d^\gamma) \rightarrow -\infty$; hence, no remote worker optimally lives near $d = 0$. Since $\{d = 0\}$ is measure zero, these limits do not affect any aggregate.

³⁰All calculations are reported in Supplemental Appendix E.E1.

Naturally, the term $\bar{\varepsilon}_{\ell\ell',t}(L_{c,t+1})$ denotes the average value of switching from ℓ to ℓ' and is given by

$$(18) \quad \bar{\varepsilon}_{\ell\ell',t}(L_{c,t+1}) \equiv \beta[V_{\ell'}(L_{c,t+1}; \omega^{t+1}) - V_{\ell}(L_{c,t+1}; \omega^{t+1})] - (F_{\ell\ell',t} - F_{\ell\ell,t}).$$

We are ready to define a dynamic equilibrium as well as a stationary equilibrium.

DEFINITION 2: *Given a known exogenous sequence of parameters $\{\omega^t\}_{t=0}^{+\infty}$ and an initial condition $0 \leq \bar{L}_{c0} \leq L$, a dynamic equilibrium is a pair of value functions $V_{\ell}, \ell \in \{c, m\}$ and a matrix of transition probabilities $\lambda_{\ell\ell'}, \ell, \ell' \in \{c, m\}$ such that equations (8) and (17) hold, and the state of the economy evolves according to (16), with $L_{c0} = \bar{L}_{c0}$.*

DEFINITION 3: *Given a time-invariant sequence of exogenous parameters $\omega \equiv \{\bar{\omega}\}_{t=0}^{+\infty}$, a stationary equilibrium is a dynamic equilibrium where all endogenous variables are constant over time.*

Throughout the text, we focus on multiplicity of stationary equilibria. In principle, the model might feature multiple equilibria, something we have not fully explored theoretically. In our numerical exercises, we have not found indications of the presence of such multiplicity.

Characterization of Stationary Equilibria.—To discuss the existence and other properties of stationary equilibria, we specify the shape of $B(\cdot)$ and $\tau(\cdot)$ and ensure that the model does not deliver corner solutions. In the rest of the paper, we maintain the following two assumptions.

ASSUMPTION 1: $B(\cdot)$ and $\tau(\cdot)$ are isoelastic functions:

$$B(\tilde{L}_c) \equiv \bar{B} \tilde{L}_c^{\xi} \text{ and } \tau(\tilde{L}_c) = \bar{\tau} \tilde{L}_c^{\theta}, \text{ with } \bar{B}, \bar{\tau}, \xi, \theta > 0.$$

ASSUMPTION 2: *The dispersion in the idiosyncratic shocks is large enough:*

$$s > \mu^2 \left(\xi + \delta - \theta - \frac{\gamma}{2\mu} \right).$$

Assumption 1 allows a characterization of stationary equilibria in terms of comparisons among (constant) elasticity values. Assumption 2 guarantees that as an economy approaches a purely remote equilibrium, the average value of switching from commuting to remote goes to $-\infty$.

The elasticity ξ captures how the CBD amenity value rises with the mass of in-office workers, while θ captures how congestion costs increase with that same activity. These parameters summarize the nonpecuniary externalities of density and operate alongside the production elasticity δ and the commuting-cost elasticity γ .

PROPOSITION 1: *There always exists at least one stationary equilibrium.*

PROOF:

See Supplemental Appendix E.E2.

The proof makes a formal case that, under Assumptions 1 and 2, any stationary equilibrium is always interior regardless of the switching costs. On one hand, the average benefit of switching to commuting approaches $+\infty$ when almost everybody is working remotely, which causes the number of commuters to stay above zero. On the other, when almost all employees commute, the number of remote workers tends to stay above zero. Since the law of motion for L_c is continuous, there must be at least one level of current commuters for which the number of commuters in the following period remains constant.

Having established existence, we now investigate the properties of the set of stationary equilibria in this economy. Proposition 2 establishes that if agglomeration forces are weak enough, there is a unique stationary equilibrium.

PROPOSITION 2: *Assume $\delta + \xi < \theta + \gamma/(2\mu)$. Then there is a unique stationary equilibrium. The stationary equilibrium value of the commuting population L_{css} monotonically increases with A and $\bar{B}$ and monotonically falls with Z , T , and $\bar{\tau}$.*

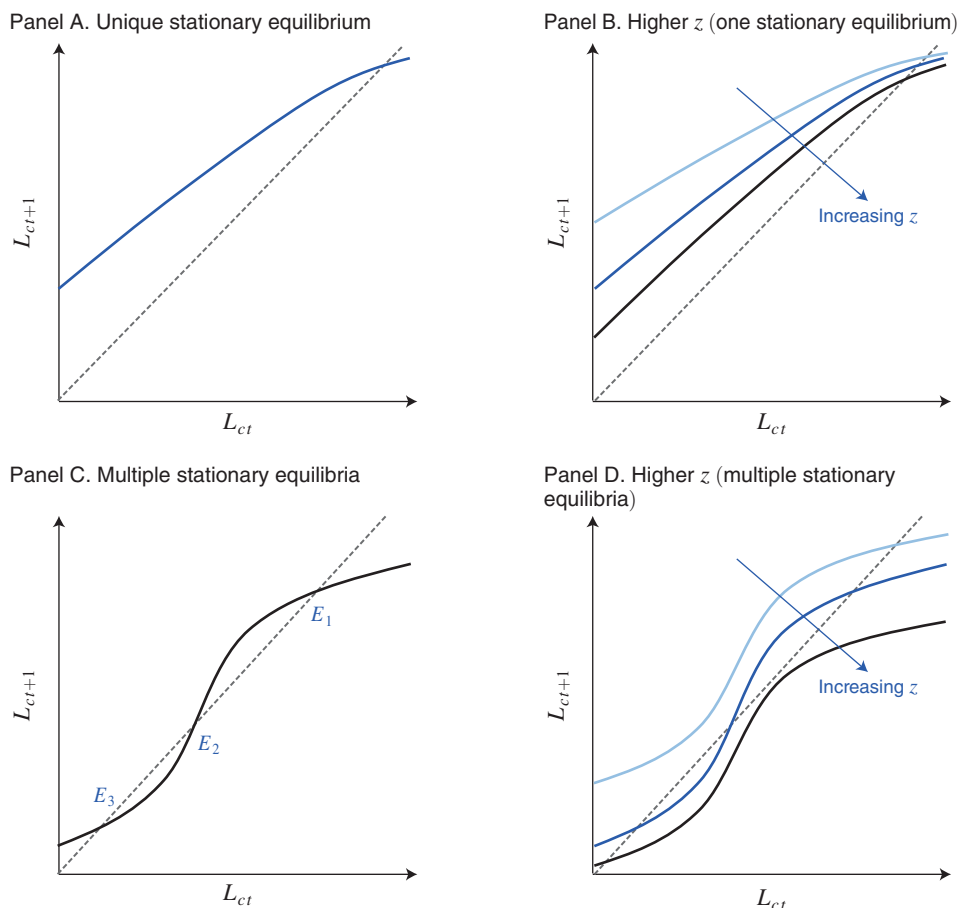
PROOF:

See Supplemental Appendix E.E3.

A unique stationary equilibrium requires that having more commuters increases the individual incentive to become a remote worker. We can break down this individual incentive into two parts: the current benefit of working remotely instead of commuting and the option value of becoming a remote worker. If the measure of commuters increases, the option value of becoming a remote worker always grows—a dynamic congestion force. Under the assumptions of Proposition 2, static congestion forces dominate agglomeration forces, and so the current net benefit, $u_m - u_c$, from being a remote worker also increases with the number of commuters. This is sufficient for uniqueness. The top left panel of Figure 7 depicts a typical phase diagram for this case.

The blue line indicates how the measure of commuters changes from one period t (horizontal axis) to the next $t + 1$ (vertical axis). The city's unique stationary equilibrium occurs at the crossing with the dashed 45-degree line. A city below the stationary equilibrium value will experience an increase in the number of commuters over time. However, congestion forces will eventually slow this growth as the number of commuters rises.

Increases in office productivity A and amenity benefits $\bar{B}$ reduce the static gains of remote work, lowering the share of commuters who switch to remote work. These changes would move the blue line up and to the left, increasing the number of steady-state commuters. Conversely, improvements in the efficiency of remote work z or its amenity benefit, T , or higher commuting frictions $\bar{\tau}$ increase the motivation to move to remote work, pushing the line down and to the right and reducing the equilibrium number of commuters. Panel B of Figure 7 shows the consequences of a progressive improvement in remote work productivity. This is reminiscent of the gradual and modest adjustment in the number of commuters before the pandemic, likely in response to advances in information and communication technology.

FIGURE 7. PHASE DIAGRAMS AND INCREASES IN THE REMOTE WORK PRODUCTIVITY z

Notes: The top panels show the case of a city with a unique stationary equilibrium; panel A depicts a typical phase diagram, and panel B shows how the phase diagram shifts in response to increases in remote work productivity, from the light blue line to the black line. The bottom panels of the figure show the case of a city with multiple stationary equilibria; panel C depicts a typical phase diagram, and panel D indicates how its phase diagram shifts in response to increases in remote work productivity, from the light blue line to the black line. In all panels, the dashed gray line is the 45-degree line.

The discussion above suggests that multiplicity can arise when an increase in the number of commuters strengthens the incentive to become a commuter. These conditions occur when static agglomeration forces overcome dynamic congestion forces.

The bottom left panel of Figure 7 illustrates a typical phase diagram for this scenario with three stationary equilibria, E_1 , E_2 , and E_3 . If a city is located between E_1 and E_2 , the number of commuters increases toward E_1 . However, if a city is situated between E_2 and E_3 , the number of commuters decreases until it reaches the stationary equilibrium E_3 . Thus, E_2 is an unstable stationary equilibrium, and any perturbation away from it will cause the city to depart from that equilibrium.

Perhaps surprisingly, strong agglomeration forces alone are not sufficient for multiplicity. The bottom right panel of Figure 7 shows that the phase diagram shifts down and to the right, from lighter to darker colors, as remote work efficiency improves. If remote work is not efficient, having fewer commuters does not increase the relative incentive to work from home enough to induce more workers to work remotely. Conversely, if remote work is highly efficient, the “high-commuting” equilibrium vanishes as the incentive to switch to remote work is consistently strong. Proposition 3 makes this intuition formal and characterizes the “cone of multiplicity,” namely, the set of parameters for which we obtain multiplicity of stationary equilibria.

PROPOSITION 3: *Suppose $F_{cc} = F_{mm} = 0$. Then, there exist a finite threshold $\eta_{min} > \theta + \gamma/(2\mu)$ and a set $\mathcal{Z} \subset \mathbb{R}_{++}$ such that:*

- (i) *for $\delta + \xi > \eta_{min}$, $\mathcal{Z}$ is a nonempty interval (Z_{min}, Z_{max}) , and there are multiple stationary equilibria if $z/A \in \mathcal{Z}$; further, Z_{min} and Z_{max} are functions of all the parameters of the model, they grow with L and $\bar{B}$, and fall with T and $\bar{\tau}$;*
- (ii) *there is a unique stationary equilibrium in all other cases.*

PROOF:

See Supplemental Appendix E.E4.

The left panel of Figure 8 provides a visual representation of this proposition.

The figure displays the combinations of $(\delta + \xi, z/A)$ for which a city exhibits multiple stationary equilibria, holding the remaining parameters constant. When $\delta + \xi < \theta + \gamma/(2\mu)$, no values of z/A can generate multiple stationary equilibria. This is essentially a restatement of Proposition 2.³¹ If agglomeration forces are high enough ($\delta + \xi > \eta_{min} > \theta + \gamma/(2\mu)$), there is a range of z/A where the city exhibits multiplicity. Intuitively, η_{min} summarizes the minimum level of static and dynamic congestion forces that agglomeration forces need to overcome for multiple stationary equilibria to be possible, although not guaranteed; as in panel D of Figure 7, the efficiency of remote work can neither be too low nor too high.

An implication of Proposition 3 is that for some values of z/A , increasing agglomeration forces will change the structure of the city from a single to multiple stationary equilibria. This fact is also easily seen in Figure 8. In the left panel, we consider a movement from point P_1 , outside the cone of multiplicity, toward P_2 , inside the cone. We know this movement can be generically constructed because $\mathcal{Z}$ is nonempty for at least some values of $\delta + \xi$. This comparison corresponds to the movement from the phase diagram F_1 to F_2 in the right panel. An increase in the

³¹ Proposition 3 imposes $F_{cc} = F_{mm} = 0$, a slightly more restrictive assumption. Without it, $\mathcal{Z}$ need not be an interval for low values of $\delta + \xi$. Instead, there are two thresholds, η_{min} and $\eta_{max} \geq \eta_{min}$; $\mathcal{Z}$ is generally the union of disjoint intervals, and it is exactly one interval only when $\delta + \xi > \eta_{max}$. We believe it is reasonable to assume no fixed costs to remain in the same labor delivery mode. Therefore, we focus on this case in the main text.

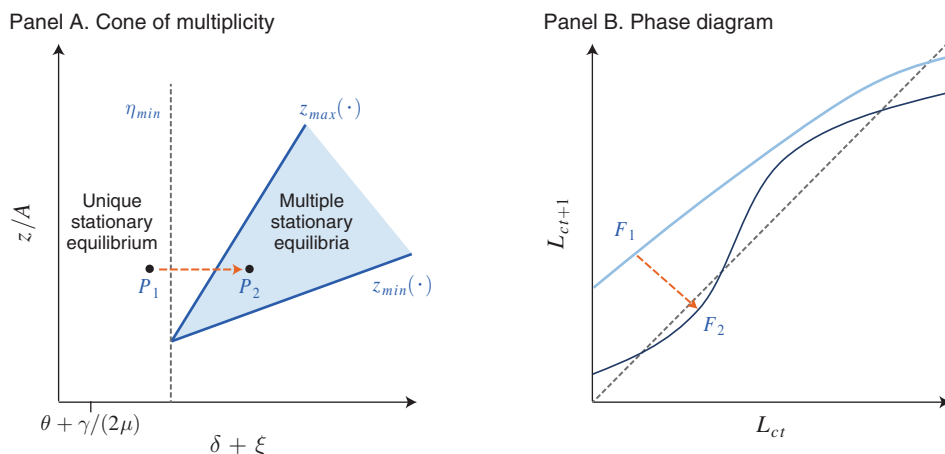


FIGURE 8. IMPLICATION OF HIGHER AGGLOMERATION FORCES

Notes: The left panel shows the combinations of z/A and $\delta + \xi$ for which, given the remaining parameters, the city displays multiple stationary equilibria. Changes in the remaining parameters move the cone. The dashed gray line indicates the minimum value of agglomeration externalities that generate multiple stationary equilibria. The left panel also shows that for some values of z/A , an increase in $\delta + \xi$ can switch the city from exhibiting a unique (P_1) to multiple (P_2) stationary equilibria. The right panel displays how the phase diagram changes from F_1 to F_2 in association with this switch. The dashed gray line in the right panel shows the 45-degree line.

strength of agglomeration forces exacerbates the role of coordination mechanisms and can create multiple stationary equilibria.

Because our framework treats each city as a closed economy, cities with strong agglomeration externalities cannot expand by attracting additional workers. This does not imply, however, that in a world with many cities the city with strongest agglomeration would attract the entire population in equilibrium. The parameters governing agglomeration are elasticities; as a high-agglomeration city grows, each additional worker represents a smaller proportional increase in population and thus a diminishing marginal effect on productivity and amenities. Moreover, idiosyncratic preferences for city attributes, together with congestion and migration frictions across locations, can ensure a spatially dispersed population even when some cities operate under locally increasing returns.

These results can speak to the organization of production in cities both in the long and short run. In the long run, secular trends alter the position of a city with respect to its cone. For example, the trends in remote work discussed in Section I suggest that a city with a high enough $\delta + \xi$ may traverse its cone from a single equilibrium to a region of multiplicity and suddenly lose the high-commuting equilibrium in favor of the low-commuting one. From a shorter-run perspective, lockdown policies during the recent COVID-19 pandemic can be seen as a negative exogenous shock to the number of commuters. Our theory envisions heterogeneous adjustments across cities, depending on whether they find themselves inside or outside their cone when the shock occurs. We now turn to the estimation of the parameters of our model. After quantifying our model for most US cities in Section IV, we relate the predictions of our model to the evidence in Section V.

IV. Quantification

When cities can potentially feature multiple equilibria, quantifying the model requires either taking a stand on which equilibrium the data is a realization of or relying on relationships that hold in any equilibrium. For this reason, we rely entirely on data that were produced before 2019, a period during which every city was arguably in the high-commuting equilibrium. More importantly, we only use predictions of the model that do not depend on which equilibrium—stationary or not—is being realized.

We quantify our model by estimating parameters that are specific to each Core-based Statistical Area in the United States when possible. We proceed in four steps. First, we estimate the dispersion of idiosyncratic preferences, which is inversely related to the elasticity of transitions into remote work with respect to earnings and the fixed cost of transitioning between delivery modes. Second, we construct estimates of the agglomeration externality in production. Third, we estimate the parameters of the remote work technology. Fourth, we estimate the transportation costs elasticity from city-specific measures of housing rent distance gradients.

A. Estimation of the Transition Elasticity and Transition Costs

We start with the identification of the variance of idiosyncratic preference shocks for labor delivery modes, the fixed costs, and a time trend for the amenity value of remote work. As in Proposition 3, we assume $F_{cc} = F_{mm} = 0$. We further assume that $F_{cm} = F_{mc} = F$, reducing the quantification of fixed transition costs to the identification of F . Following the work of Artuç, Chaudhuri, and McLaren (2010) on the estimation of discrete choice models, and of Arcidiacono and Miller (2011) in the structural labor literature on conditional choice probabilities (CCPs), we exploit the log-linear relationship between observed transition probabilities across labor delivery modes and workers' payoffs, including transition costs, income, and continuation values. Through a linear regression, the elasticity of transitions between labor delivery modes is identified from the responsiveness of transitions to income changes, while fixed transition costs are additionally identified from the overall level of transitions.³² We use the following result, which provides an expression of forward-looking relative transition probabilities as a function of the within-period flows of utility, transition costs F , and transition elasticity s only.

LEMMA 1: Suppose $F_{cc} = F_{mm} = 0$ and $F_{cm} = F_{mc} = F$. Then

$$(19) \quad y_{\ell\ell't} \equiv \ln \frac{\lambda_{\ell\ell't} \lambda_{\ell'\ell'+t+1}^\beta}{\lambda_{\ell\ell't} \lambda_{\ell'\ell'+t+1}^\beta} = \frac{\beta}{s} (u_{\ell't+1} - u_{\ell'+t+1}) - \frac{(1 - \beta)F}{s}.$$

PROOF:

See Supplemental Appendix E.E5.

³² See Traiberman (2019) for an application to occupational and sectoral transitions.

Intuitively, a sluggish response of transitions to differences in utility flows suggest that a large variance in idiosyncratic shocks (high s) is overshadowing the significance of those differences; conditional on a variance, high average gross flows are then indicative of small switching costs (low F) between states. Technically, this approach considers one-period deviations from an otherwise common sequence of labor delivery modes to difference out continuation values. We can write the difference in the utility flows from (15) as

$$(20) \quad u_{mt+1} - u_{ct+1} = \ln \frac{w_{mt+1}}{w_{ct+1}} + \mu \ln T_{t+1} - \mu \ln B(\tilde{L}_{ct+1}) \\ + \mu \ln \tau(\tilde{L}_{ct+1}) + \frac{\gamma \mu}{2} \ln \frac{L_{ct+1}}{\pi}.$$

In order to obtain a specification that can be estimated in a linear regression, we maintain Assumption 1 about the parametric specification of the city-center amenity $B(\tilde{L}_c)$ and congestion $\tau(\tilde{L}_c)$ and assume that the remote work amenity takes the form of a log-linear function of time. That is, we let $T_t = \exp\{\alpha_0 + \alpha_1 t\}$. Our estimating equation can then be written as

$$(21) \quad y_{\ell\ell'} = \eta_0 \cdot \ln \frac{w_{\ell't+1}}{w_{\ell't+1}} + \eta_1 + d_{\ell\ell'} \cdot (\eta_2 + \eta_3 \cdot (t+1) + \eta_4 \cdot \ln L_{ct+1} \\ + \eta_5 \cdot \ln L_{t+1}) + \hat{\varepsilon}_{\ell\ell'},$$

where $d_{\ell\ell'} = 1$ is a discrete variable taking the value 1 if the transition is from commuting to remote, that is, if $\ell\ell' = cm$ and takes the value -1 if the transition is from remote to commuting, that is, $d_{\ell\ell'} = -1$ if $\ell\ell' = mc$, and $\hat{\varepsilon}_{\ell\ell'}$ is an expectation error.

With this specification, the parameters s , F , and $\xi - \theta$ can be recovered from the estimated coefficients according to

$$(22) \quad s = \frac{\beta}{\eta_0}, F = -\frac{\beta}{1 - \beta} \frac{\eta_1}{\eta_0}, \xi - \theta = -\frac{1}{\mu(1 - \mu)} \frac{\eta_5}{\eta_0}.$$

In principle, we should compute $y_{\ell\ell'}$ for each city. However, the NLSY sample does not contain enough observations of these transitions to obtain them accurately for each city. This is especially true if we define remote work as spending three days per week or more at home, a choice that was rare before the COVID-19 pandemic.³³ For this reason, we impose additional assumptions to implement the estimation of (21). In our main specification, we partition cities based on four US regions, the most precise geographic units recorded in the NLSY public-use data.³⁴ We define transition shares for each of these four regions and for two groups of educational attainment (with or without a college degree). That is, for each year, we construct the empirical equivalent of $y_{\ell\ell'}$ in (21) for eight region-education groups, indexed

³³ Out of the 4,847 individuals observed between 2000 and 2018 in our sample, only 386 ever reported working remotely for 3 days per week, while 650 and 1,074 worked remotely 2 days or more per week, and 1 day or more per week, in at least one of the survey years, respectively.

³⁴ The four regions are Northeast, North-Central, South, and West.

by r . We define a remote worker as someone who spends at least one day per week at home to maximize the number of observed transitions.³⁵

The construction of $y_{\ell\ell't}^r$ for each group r is based on the fraction of individuals $\lambda_{\ell\ell't}^r$ who make a transition to or from remote work between two consecutive periods of time. To compute these shares for each group r , we first control for the role of covariates other than education and region in driving the transition shares of each group that are not directly captured by our model. These include differences in the industrial and occupational composition across groups that could drive differences in transition shares. Specifically, we estimate a probit model for the share of remote workers, controlling for year-group r, t indicators, the role of age and experience, and occupation and industry fixed effects. We use as measures of $\lambda_{\ell\ell't}^r$ the predicted transition shares from our probit model for each group r , setting all other covariates to their mean value. We then assemble the yearly transition shares into the relative shares $y_{\ell\ell't}^r$, setting β to 0.96 on an annual basis, corresponding to a discount rate of about 4 percent.

To construct wages w_{ct}^r and w_{mt}^r , we follow a similar approach by removing the contribution of industrial and occupational determinants of the wage levels. We regress individual log hourly wages on age, tenure and their square, year-group r, t indicators, and person, industry, and occupation fixed effects. We use the residuals from these regressions to obtain a measure of wages for each year-group r, t after controlling for individual variation. The wage for a region is then the average residual wage for all the observations in a region-time.³⁶ Finally, we construct measures of total employment L_t^r and of the number of commuters L_{ct}^r in each year-group r, t from the ACS.

Our model assumes workers are forward-looking and make decisions in period t based on wages in period $t + 1$. However, if the relative wages of remote workers and commuters are imperfectly observed by workers or measured with error, OLS estimates will likely yield downward-biased estimates of η_0 . We address this concern by using past values of the wage ratio as an instrument for the future ratio.

Table C1 in Supplemental Appendix C.C1 reports the results of the estimation of (21) and the implied structural parameters. We obtain statistically significant OLS and IV estimates of η_0 and η_1 at the 1 percent level. We use the IV estimates of $\eta_0 = -0.47$ and $\eta_1 = 3.09$ to construct estimates of s and F .

We obtain an estimate of s equal to 0.30. We are not aware of existing estimates of this parameter. This value of s corresponds to an elasticity of transitions into remote work of 3.09. This elasticity can be interpreted in two slightly different ways. First, if we return to (19) and (20), it means that an increase of 1 percent in the wage of remote workers relative to the wage of commuters can induce an “acceleration” of transitions to remote work by 3.09 percent. That is, transitions into remote work would increase in period t to benefit from these higher wages—the numerator in the

³⁵ In Supplemental Appendix C.C1, we conduct an alternative exercise assuming that cities can be partitioned into subsets that offer similar remote work conditions and define remote work as spending at least two days per week at home. We then combine the decisions of individuals in cities in a given subset and form transition shares at that level. We obtain similar parameter estimates.

³⁶ This specification removes static sources of heterogeneity but preserves within-industry and within-occupation time variation in the relative compensation of remote versus in-person work. Hence, w_{ct}^r and w_{mt}^r capture changes in the productivity or returns to remote work over time rather than shifts in occupational or industrial composition.

definition of $y_{\ell\ell't}$ —relative to transitions that would have occurred in period $t + 1$ —the denominator. Alternatively, one can easily show from (17) that the elasticity of the probability of transition to remote work λ_{cmt} (relative to the probability of not transitioning λ_{cct}) with respect to the relative *value* of remote work, $V_{mt+1} - V_{ct+1}$, is also β/s . That is, if the wage of remote workers relative to the wage of commuters were to increase permanently by 1 percent, then transitions into remote work would increase by 3.09 percent in period t .

Our estimate for the transition cost F is 1.78. Given the specification of the utility of consumption in log terms, we can interpret the cost F as being equivalent to giving up $100(1 - \exp(-F)) = 83$ percent of earnings in year t in order to transition into remote work. This large number is reminiscent of the very large transition costs estimated in structural labor models.³⁷ These large estimates for the cost F reflect the fact that despite the existence of wage differentials between labor delivery modes, the mobility of workers remains quite low.³⁸ Of course, if transitioning between modes also implies changing homes in order to live at the appropriate distance to the CBD, these large fixed costs are natural.

All other coefficients are imprecisely estimated, providing no guidance on the values for ξ , θ , α_1 and the combination $\alpha_0 - \ln(\bar{B}/\bar{\tau})$. As a result, we set α_1 and $\alpha_0 - \ln(\bar{B}/\bar{\tau})$ to zero. In Section IVE, we calibrate $\bar{\tau}$ for each city to match the initial value of L_c/L , and set ξ and θ to values consistent with the literature.

B. Agglomeration Externalities

We estimate the agglomeration externality δ_j for each CBSA j , where δ_j measures the percentage increase in labor value added associated with a 1 percent increase in the number of commuters to the CBD in city j .

Our estimation strategy builds on the wage expression in equation (3), augmented to allow for time-varying in-office productivity A_{jt} . The wage of a commuter in CBSA j is therefore

$$(23) \quad w_{cjt} = A_{jt} \tilde{L}_{cjt}^{\delta_j}$$

Each CBSA's in-office productivity A_{jt} may depend on an external factor K_{jt} (e.g., the quality of its business environment, transit network, or other relevant infrastructure) that can itself scale with total employment. Let $A_{jt} = \bar{A}_{jt} K_{jt}^{\nu_j}$ and $K_{jt} = g_{jt} L_{jt}^{\vartheta_j}$, so that $A_{jt} = \bar{A}_{jt} g_{jt} L_{jt}^{\varsigma_j}$, where $\varsigma_j \equiv \nu_j \vartheta_j$ governs the elasticity of in-office productivity with respect to city size.³⁹ We can then decompose in-person log productivity as

$$\log A_{jt} = \tilde{a}_t + \tilde{b}_j t + \tilde{c}_j + \varsigma_j \log L_{jt} + \tilde{u}_{jt}.$$

³⁷ Artuç, Chaudhuri, and McLaren (2010) estimate switching costs between occupations that are equal to at least four times annual wages. In their study of location decisions, Kennan and Walker (2011) estimate moving costs amounting to US\$(2010)312,000 on average.

³⁸ In Supplemental Appendix C.C1, we estimate s and F with the restricted-use version of the NLSY geo-code data from the US Bureau of Labor Statistics (2020a) that is geocoded at the county level. We obtain similar estimates.

³⁹ See Supplemental Appendix C.C2 for the derivation of the wage function and the estimation equation (25) from a production function featuring K_{jt} as an explicit input.

Defining $\Delta x_t = x_t - x_{t-\Delta t}$ as the Δt -period change in variable x , the implied change in productivity over Δt periods is

$$(24) \quad \Delta \log A_{jt} = a_t + b_j + \varsigma_j \Delta \log L_{jt} + u_{jt},$$

where $a_t \equiv \Delta \tilde{a}_t$, $b_j \equiv \tilde{b}_j \Delta t$, and $u_{jt} = \Delta \tilde{u}_{jt}$.

Let $S_{cjt} \equiv L_{cjt}/L_{jt}$ denote the share of commuters. Combining equations (23) and (24) with the definition of $\tilde{L}_c$ in equation (2) yields

$$(25) \quad \Delta \log w_{cjt} = a_t + b_j + \mu \delta_j \Delta \log S_{cjt} + (\delta_j + \varsigma_j) \Delta \log L_{jt} + u_{jt}.$$

Equation (25) forms the basis of our estimation strategy. It implies that commuter wages vary more with the share of commuters in cities with strong agglomeration externalities. Importantly, it also shows that wage changes associated with city size L_{jt} may reflect additional channels beyond the in-person externality δ_j . Hence, to estimate δ_j , we focus on identifying the causal effect of changes in commuter shares rather than population changes.

Specifically, our empirical strategy proceeds in three steps. First, we define the empirical counterpart of commuter wages in CBSA j as the average wage $\bar{w}_{cjt}$ of commuters in that CBSA, after controlling for individual characteristics (see Supplemental Appendix C.C2).

Second, we adopt an instrumental variables approach to isolate the variation in a CBSA's commuter share that is orthogonal to changes in residual productivity u_{jt} .⁴⁰ For every CBSA-year (j, t) , we predict the change in the commuter share between $t - \Delta t$ and t , labeled IV_{jt} , as a function of the CBSA's 1980 occupational composition, L_{sjt_0}/L_{jt_0} , and the economy-wide changes in occupation-specific commuter shares, $\Delta \log S_{c, st}$:

$$(26) \quad IV_{jt} = \sum_{s=1}^S \frac{L_{sjt_0}}{L_{jt_0}} \Delta \log S_{cst}.$$

Third, we assume that agglomeration forces are industry specific and that CBSAs' effective agglomeration strengths differ according to their employment composition. Following Rossi-Hansberg, Sarte, and Schwartzman (2021), we group industries into five sectors, $g = 0, \dots, 4$: Retail, Utilities, and Construction (RUC); Health and Education (HE); Professional and Other Services (PS); Manufacturing (M); and Accommodation, Trade, and Transportation (ATT).⁴¹ Denoting CBSA

⁴⁰It is not necessary for $\Delta \log L_{jt}$ to be exogenous in equation (25) for the instrumental variable strategy to yield a consistent estimate of δ_j . As shown in standard treatments of two-stage least squares (see, e.g., Wooldridge 2010, chap. 5.2.1), endogeneity of included controls does not bias the IV estimate of the coefficient on the endogenous regressor of interest, provided that the instrument is valid. We treat the role of total employment $\Delta \log L_{jt}$ as a city-specific slope without linking it to δ_j because of this potential endogeneity.

⁴¹We introduce the five industry groups for estimation tractability, allowing us to parsimoniously estimate the cross-city distribution of δ_j with only five parameters instead of one per CBSA. Introducing distinct industry groups with their own in-office and remote productivities within the model would generate heterogeneity in commuting shares across industries and, consequently, some degree of spatial sorting by industry, which we abstract from here.

TABLE 1—ESTIMATES OF CITY-SPECIFIC PARAMETERS

	Agglomeration externality δ_j	Remote work productivity z_j	Rel. remote work productivity z_j/A_j	Transport cost elasticity γ_j
Mean	0.067	16.6	2.05	0.027
10th percentile	0.040	13.5	1.43	0.002
50th percentile	0.067	15.9	1.97	0.014
90th percentile	0.092	20.3	2.82	0.044
Number of CBSAs	619	626	626	278
Mean if emp. ≥ 1 million	0.078	23.0	2.95	0.013
Mean if emp. $\leq 100,000$	0.065	15.6	1.91	0.039

Notes: The last two rows report the mean values of the parameters for (i) the group of CBSAs with employment above 1 million (consisting of 19 CBSAs) and (ii) the group of CBSAs with employment below 100,000 (consisting of 468 CBSAs). Employment is measured as the number of employed workers between 25 and 64 years old, not self-employed, listed as residents of the CBSA in the 2019 ACS. We are able to estimate the transport cost elasticity for a smaller number of CBSAs, given the data limitations on housing rents at the block group level in smaller CBSAs.

j 's labor-value-added-weighted employment share in group g in 1980 by s_{gj0} , we specify

$$(27) \quad \delta_j = \beta_0 + \sum_{g=1}^4 \beta_g s_{gj0}.$$

We then estimate

$$(28) \quad \Delta \log \bar{w}_{cjt} = a_t + b_j + \left(\beta_0 + \sum_{g=1}^4 \beta_g s_{gj0} \right) \Delta \log S_{cjt} + \gamma_j \Delta \log L_{jt} + u_{jt},$$

using $s_{gj0}IV_{jt}$, $g = 1, \dots, 4$ as instruments for each interaction $s_{gj0} \Delta \log S_{cjt}$. Table C3 in Supplemental Appendix C.C2 reports the individual coefficients. Combining the industry-specific IV estimates with each CBSA's labor-value-added-weighted employment shares yields city-specific measures of the agglomeration externalities δ_j . Summary statistics for these δ_j 's are reported in Table 1. Figure 10, panel A plots the 619 CBSA-specific estimates against employment, showing a mean of 0.067 and a standard deviation of 0.022. On average, a doubling of city size is associated with an elasticity about 0.004 points higher. Supplemental Appendix D.D4 further demonstrates that our main results are robust to calibrations using lower values of δ_j .

C. Remote Work and In-Office Technology

We also estimate in-office and remote productivities A_{jt} and z_{jt} for each CBSA j . We develop further the exercise presented in Section I by estimating the remote work premium by occupation. Using data from the NLSY, we estimate the remote work premium for every incremental hour spent working from home. Specifically, we denote by $h_{remote_{it}}$ the number of hours worked at home and predict $\hat{h} = 24$ hours per week, equivalent to three days per week of remote work.⁴²

⁴²In Supplemental Appendix C.C3, we estimate the remote work premium from the ACS. Supplemental Appendix Table C5 compares the two approaches.

We estimate a national trend and an intercept for each of the 22 occupation groups, denoted by o .⁴³ Specifically, we estimate

$$(29) \quad \ln w_{it} = (\psi_{o(i)}^0 + \psi^1 t) h_{remote_{it}} + X_{it} \beta_{o(i)} + \epsilon_{it},$$

where $\ln w_{it}$ represents the log of the hourly wage for individual i at time t ; $\psi_{o(i)}^0$ and ψ^1 represent the intercept for occupation group o and the national trend, respectively; and X_{it} is a vector of individual-level control variables, including age, age squared, tenure at the job, tenure squared, industry, marital status, race, education-year, number of children, metro area-year, and individual fixed effects. We focus on full-time US-born workers in metropolitan areas, observed between 2000 and 2018.

Finally, we estimate the remote work premium in city j using

$$(30) \quad \psi_{jt} = \sum_o sh_{oj}^{2018} (\psi_o^0 + \psi^1 t) \hat{h},$$

where sh_{oj}^{2018} is the share of workers in city j working in occupation o in 2018. Figure 9 displays the estimates $(\hat{\psi}_{o(i)}^0 + \hat{\psi}^1 t) \hat{h}$ for $t = 2018$ for each of the 22 occupation groups obtained from (29), and the estimates ψ_{j2018} of the remote work premium by CBSA obtained from (30). Both are plotted against the share of teleworkable employment estimated by Dingel and Neiman (2020).⁴⁴

From equations (29) and (30), we can interpret $(\psi_{o(i)}^0 + \psi^1 t) \hat{h}$ as an estimate of $\ln(w_{mt}/w_{ct})$ for each occupation and ψ_{jt} as an estimate of $\ln(w_{mt}/w_{ct})$ for each CBSA. We use the specification of wages in (3) to map our estimates of the remote work premium into the remote work technology parameter, z :

$$(31) \quad z_{jt} = \exp(\hat{\psi}_{jt}/\mu) w_{cjt},$$

where w_{cjt} is the average hourly wage of commuters in each CBSA j . We then recover the in-office productivity using

$$(32) \quad A_{jt} = \tilde{L}_{cjt}^{-\delta_j} w_{cjt},$$

given the city-level estimates of δ_j obtained in Section IVB. Table 1 reports summary statistics for these estimates.

D. Transportation Costs Elasticity

We estimate the transportation costs elasticity γ_j for each CBSA j . We use the relationship that relates γ_j to the slope κ_{jt} of the housing rent schedule with respect to distance to the CBD.⁴⁵ From (14), the average elasticity of rent to distance to the

⁴³ We describe these occupation groups in Supplemental Appendix A.A2.

⁴⁴ In Supplemental Appendix C.C3, we report the estimates for each occupation group and CBSA.

⁴⁵ Our model assumes that there is a fixed housing supply of one house per plot. In Supplemental Appendix C.C4, we show that a model where the number of housing units endogenously responds to the rents would deliver, for practical purposes, a very similar gradient. This result is a consequence of the number of commuters being a very high share of the overall employment before 2020.

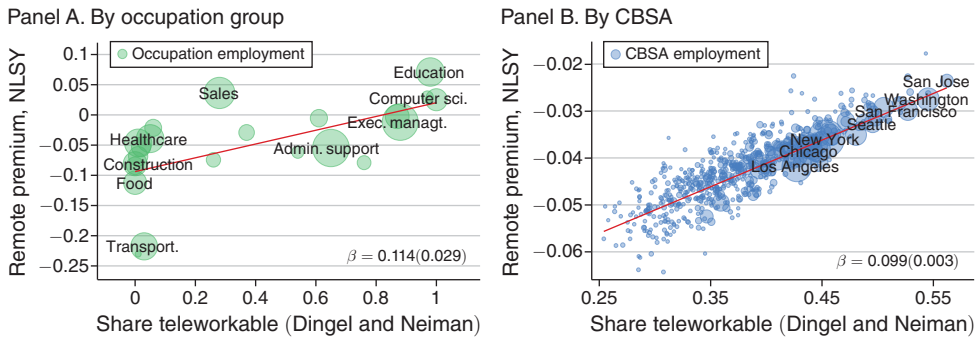


FIGURE 9. ESTIMATES OF THE REMOTE WORK PREMIUM

Notes: Panel A reports estimates for each of the 22 occupation groups obtained from (29), against the share of teleworkable employment from Table 1 in Dingel and Neiman (2020). Panel B reports the estimates of the remote work premium by CBSA obtained from (30) plotted against the share of teleworkable employment at the CBSA level. We compute the CBSA-level shares of teleworkable employment as the weighted average of the occupation-specific measures from Table 1 in Dingel and Neiman (2020). The weights are given by the CBSA share of employment of each occupation from the census sample. The coefficient β and the associated robust standard error from the regression of the remote premium on the share of teleworkable employment are reported at the bottom right of each figure.

CBD in period t is exactly related to the slope of the housing rent schedule according to

$$(33) \quad \kappa_{jt} = -\frac{\gamma_j}{1-\alpha} \left(1 - \mu + \mu \left(\frac{L_{cjt}}{L_{jt}} \right)^{1/2} \right).$$

In order to estimate κ_{jt} , we use data on housing rents at the block group level from the ACS via IPUMS NHGIS (Schroeder et al. 2025), corresponding to average values between 2015 and 2019. For each CBSA, we estimate a regression at the block group level of the log of the median housing rental price p_{ij} in each block group i on the log of the distance, $dist_{ij}$, between the centroid of block group i and the centroid of the closest CBD of CBSA j .⁴⁶ We then estimate

$$(34) \quad \ln p_{ij} = \kappa_j \ln(dist_{ij}) + \beta_j X_{ij} + \alpha_j + e_{ij},$$

where the controls X_{ij} contain block-level measures of the percentages of Hispanic, Black, and Asian population; dwelling characteristics such as the percentage of dwellings by type of structure, by the number of bedrooms, and by construction decade; as well as other geographic controls defined at the tract level and obtained from Lee and Lin (2018).⁴⁷ The summary statistics of the estimates of γ_j across

⁴⁶ We provide details on how we identify each CBSA's CBDs in Supplemental Appendix A.A3.

⁴⁷ Geographic controls include the log of land area in square kilometers, number of housing units, the average age of housing units, average terrain slope, the maximum average temperature in July, the minimum average temperature in January, annual precipitations, log of distance to the closest river, to the closest lake, and to the closest shore.

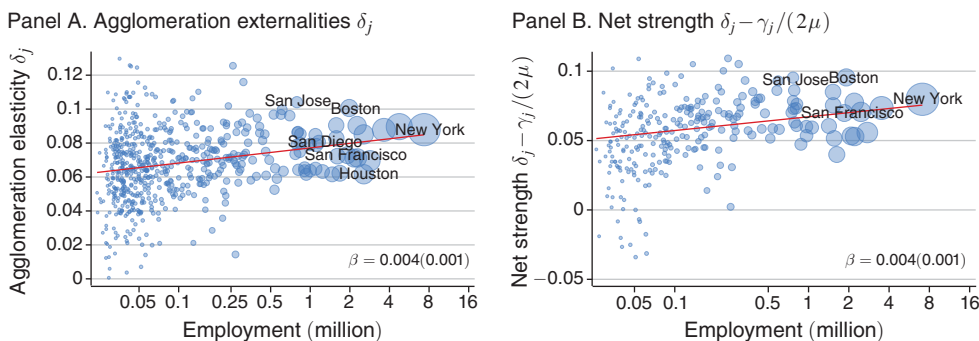


FIGURE 10. AGGLOMERATION EXTERNALITIES AND CBSA SIZE

Notes: Panel A reports the city-specific δ_j against the CBSAs' BEA employment. Each δ_j is estimated from equation (28). The size of the circles represents CBSAs' employment. Panel B reports the city-specific $\delta_j - \gamma_j / (2\mu)$ against the CBSAs' BEA employment.

CBSAs are reported in Table 1, and Supplemental Appendix C.C4 shows their relation with city size.⁴⁸

Although the estimated values of γ_j are lower than those found in studies such as Duranton and Puga (2023), this difference reflects model structure rather than data discrepancies. In our framework, where housing is a consumption good, the mapping from the observed rent gradient to the transportation-cost elasticity is scaled by the share of nonhousing consumption, $1 - \alpha$. In contrast, models with fixed housing consumption identify the rent gradient directly as γ . As shown in Supplemental Appendix C.C4, this distinction mechanically results in smaller implied elasticities for the same empirical slope. Supplemental Appendix D.D4 further demonstrates that our main results are robust to calibrations using higher values of γ_j .

Proposition 3 indicates that the resulting net strength of the agglomeration externalities, $\delta_j - \gamma_j / (2\mu)$, is a crucial statistic for the potential of a city to exhibit multiple stationary equilibria. In Figure 10, panel B, we plot such net strength against the CBSAs' employment size. We find that this net strength is positively related to city size; a doubling in city size is associated to an increase of 0.004 of this net strength, out of a mean value of 0.049.

E. Quantification of Remaining Parameters

As discussed in Section IVA, our attempt to estimate the difference between ξ and θ yields statistically insignificant results that are not informative. We therefore calibrate these parameters using estimates from the literature. We set $\xi = 0.15$, consistent with evidence on residential amenity elasticities from Glaeser, Kolko, and Saiz (2001); Ahlfeldt, Redding, Sturm, and Wolf (2015); and the synthesis in

⁴⁸For small CBSAs, the number of block groups with nonmissing observations on rents and other control variables is sometimes small, leading to estimates of γ_j that are not statistically different from 0 at the 10 percent level. We drop those cities due to the lack of data.

Redding (2022).⁴⁹ We similarly set $\theta = 0.15$, in line with congestion estimates ranging from 0.13 to 0.17 in Couture, Duranton, and Turner (2018); Akbar, Couture, Duranton, and Storeygard (2023); and Almagro et al. (2024).⁵⁰ As a result, our baseline calibration imposes $\xi - \theta = 0$. Supplemental Appendix D.D5 shows that our main results are qualitatively robust to varying $\xi - \theta$ in the empirically relevant range between -0.03 and 0.03 .

We calibrate $\bar{\tau}_j$ for each city to match its observed value of L_{jc}/L_j . As shown in Supplemental Appendix E.E2, a stationary equilibrium value for L_{jc}/L_j can be expressed in terms of λ_{cc} . Hence, we assume that the city is in the high-commuting stationary equilibrium in 2019, infer the value of λ_{jc} that is consistent with L_{jc}/L_j , and then solve for the value of $\bar{\tau}_j$ that makes such λ_{jc} a stationary equilibrium.

We set the share of consumption expenditures on tradable goods to $\alpha = 0.76$, following Diamond and Moretti (2024) and Davis and Ortalo-Magné (2011). The share of days per week spent at home for remote workers is set to $\mu = 3/5$.

V. Permanent Responses to Transitory Shocks

The analysis in Section IIID indicates that a city in the cone of multiplicity could permanently switch to a path that converges toward the low-commuting stationary equilibrium when subject to a large enough shock to the number of commuters. In this new stationary equilibrium, trips to the CBD are permanently lower, and the rent and value gradients are permanently flatter. In our data, the response heterogeneity as a function of city size of Figures 3 and 5 appears to have stabilized by 2022. To interpret this evidence, we turn to a quantified version of our model. We use city-specific parameters retrieved in Section IV to calculate the values of z_j/A_j , $\delta_j - \gamma_j/(2\mu)$, and the corresponding set $\mathcal{Z}_j$ defined in Proposition 3. We construct these values for 278 CBSAs with valid information on remote work premiums and elasticities.⁵¹ For each city, we create an indicator variable $\mathbf{1}_j^{\text{cone}} = 1$ if $z_j/A_j \in \mathcal{Z}_j$, and zero otherwise, meaning the city is inside its cone of multiplicity. We find that 208 cities were in their respective cone before the pandemic. These cities could be pushed into a low-commuting equilibrium by a large enough, albeit temporary, shock to the number of commuters.

Our theory appears to have explanatory power. We estimate a city-level probit of $\mathbf{1}_j^{\text{cone}}$ on the July 2022 measure of CBD trips, relative to January 2020. This regression returns a negative and significant coefficient: Cities with a lower level of trips postpandemic were indeed much more likely to be in their cone before the pandemic. Figure 11, panel A reports this estimated share of cities that are inside

⁴⁹Glaeser, Kolko, and Saiz (2001) interpret the stronger sensitivity of housing prices than wages to city size as evidence of demand for urban amenities, implying an amenity elasticity of about 0.14 for US cities in 1990. Using within-city spatial variation, Ahlfeldt, Redding, Sturm, and Wolf (2015) estimate a residential amenity elasticity of approximately 0.15. Redding (2022) reviews this evidence and concludes that an amenity elasticity around 0.15 is consistent across a range of urban settings.

⁵⁰The closest empirical counterparts to congestion externalities in our model are estimates of how additional road users reduce average travel speed. Using city-level data for India, Akbar, Couture, Duranton, and Storeygard (2023) estimate a population elasticity of travel speed of about -0.17 . Using US metropolitan data, Couture, Duranton, and Turner (2018) estimate an elasticity of travel speed with respect to aggregate travel of about -0.13 , while Almagro et al. (2024) report a preferred elasticity of travel time with respect to traffic flows of about 0.17 for cars.

⁵¹The model assumes positive δ and γ and requires a remote work wage discount for a valid calibration.

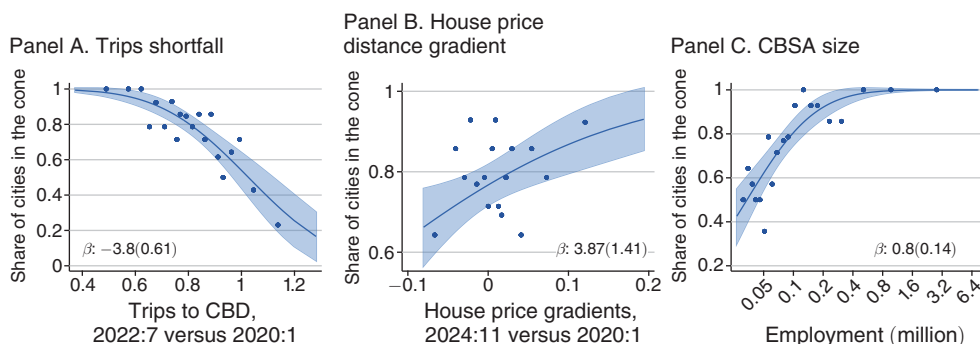


FIGURE 11. SHARE OF CITIES IN THE CONE OF MULTIPLICITY BY CBSA CHARACTERISTICS

Notes: The blue line in this figure plots the predicted share of cities in their cone of multiplicity from a probit regression of the indicator variable $\mathbf{1}_j^{cone}$ on the volume of trips in July 2022, relative to their value in January 2020 (panel A), the change in the house price distance gradients between January 2020 and November 2024 (panel B), and on the log total employment of a city (panel C), with the shaded area representing the 95 percent confidence interval. The figure also reports the coefficient and robust standard error of the independent variable in the probit regression. One blue dot is the average value of $\mathbf{1}_j^{cone}$ in each of 20 quantiles of the trip shares.

the cone as a function of the trip shortfall, while the blue dots report the fraction of cities in the cone in each of 20 bins of the independent variable.

The propensity of being inside the cone is also associated with the observed flattening of the housing price gradient. As Figure 11, panel B shows, a probit regression of $\mathbf{1}_j^{cone}$ on the change in the bid-rent housing gradient between January 2020 and November 2024 yields a positive and significant coefficient. The blue dots report the fraction of cities in the cone in each of the 20 bins of gradient change. Hence, cities that display a large persistent flattening of the house price gradient are more likely to be in their cone of multiplicity.⁵²

Overall, we find that larger cities are more likely to lie inside their cone. A probit regression of $\mathbf{1}_j^{cone}$ on log city employment yields a positive and statistically significant coefficient. Figure 11, panel C plots the implied probability as a function of log employment, together with a 95 percent robust confidence interval; the blue dots report the fraction of cities in the cone within each of 20 employment-size bins. Consistent with the fact that larger cities in the data are more likely to exhibit permanent shifts in commuting, they are also more likely to fall within the cone of multiplicity predicted by the theory.

Supplemental Appendix D.D1 shows that the probability of being in the cone increases with δ_j and z_j/A_j and decreases with γ_j , as implied by the model. We also show that these results are robust to lowering the average value of δ and increasing

⁵²Note that some small cities are missing from the analysis in Figure 11, panel A and Figure 11, panel B since there are not enough data to estimate the full set of parameters required to determine if they were in the cone of multiplicity before the pandemic. However, in general, small cities are much more likely to be outside of the cone of multiplicity, and they mostly have gone back to prepandemic commuting levels, as shown in Figure 3, panel A and Figure 5, panel A. Hence, Figure 11, panel A and Figure 11, panel B would probably show an even starker relationship if we could include more small cities.

TABLE 2—DETERMINANTS OF CHANGES IN TRIPS TO CBD: OBSERVABLE CHARACTERISTICS

	Trip shortfall			$\mathbf{1}_j^{\text{back}(5,20)}$		
	(1)	(2)	(3)	(4)	(5)	(6)
$\mathbf{1}_j^{\text{cone}}$	-14.17 (0.00)	-7.97 (0.00)	-8.00 (0.00)	-1.26 (0.00)	-1.16 (0.00)	-1.10 (0.01)
CBSA employment			-3.38 (0.00)			-6.59 (0.01)
Share of teleworkable employment		-36.75 (0.05)	-38.35 (0.03)		-5.60 (0.08)	-6.07 (0.04)
Share of trips to CBD in 2019		-0.90 (0.55)	-1.10 (0.46)		-0.29 (0.31)	-0.59 (0.03)
s_{j0} Health and education		-4.32 (0.92)	9.41 (0.83)		5.87 (0.47)	8.50 (0.31)
s_{j0} Professional services		-288.26 (0.00)	-204.95 (0.00)		-17.26 (0.10)	10.66 (0.26)
s_{j0} Manufacturing		82.35 (0.39)	109.94 (0.24)		39.63 (0.04)	52.73 (0.01)
s_{j0} Accommod., trade, transport.		-83.38 (0.41)	-47.96 (0.63)		4.75 (0.72)	14.07 (0.32)
Constant	-8.40 (0.00)	31.16 (0.05)	20.29 (0.21)	0.37 (0.07)	2.31 (0.36)	-2.61 (0.30)
Observations	275	275	275	188	188	188
R^2	0.144	0.422	0.456			
Pseudo R^2				0.137	0.392	0.494

Notes: Columns 1–3 report OLS estimates of the percentage drop in CBD trips from January 2020 to July 2022 on $\mathbf{1}_j^{\text{cone}}$ and the listed controls. Columns 4–6 report probit estimates for $\mathbf{1}_j^{\text{back}(5,20)}$. Robust p -values in parentheses.

the average value of γ (Supplemental Appendix D.D4), as well as to varying the calibrated value of $\xi - \theta$ (Supplemental Appendix D.D5).

Our explanation of the postpandemic empirical patterns rests on an indicator, $\mathbf{1}_j^{\text{cone}}$, which summarizes all the relevant aspects of a city's propensity to exhibit multiple equilibria. A natural question is whether this summary measure continues to predict postpandemic behavior once we control for other observable determinants of the return to office—even though those same observables partly shape $\mathbf{1}_j^{\text{cone}}$ through their influence on the underlying model parameters.

Table 2 shows that $\mathbf{1}_j^{\text{cone}}$ retains strong explanatory power after accounting for alternative city characteristics. The dependent variable in columns 1–3 is the percentage change in trips to the CBD in July 2022 relative to January 2020. Column 1 confirms the strong association between $\mathbf{1}_j^{\text{cone}}$ and trip shortfalls illustrated in Figure 11, panel A; the R^2 is 0.144 when including this variable alone. Columns 2 and 3 add prepandemic controls—the share of employment in teleworkable occupations, the prepandemic share of trips to the CBD (a proxy for congestion), and value added employment shares s_{gj0} in the four industry groups used to estimate agglomeration elasticities in Section IVB. The cone indicator remains highly significant. Adding total CBSA employment as an additional control leaves its coefficient virtually unchanged, confirming that the effect is not merely capturing city size.

To isolate the extensive margin of switching, we also construct a binary indicator, $\mathbf{1}_j^{\text{back}(5,20)}$, equal to 1 if trips in July 2022 were within 5 percent of prepandemic

TABLE 3—PERCENTAGE WELFARE DIFFERENCE BETWEEN THE HIGH- AND LOW-COMMUTING EQUILIBRIUM

	Min	p10	p25	p50	p75	p90	Max	Mean	N
Welfare loss $(-\hat{\mathcal{W}})$	1.2	1.6	1.9	2.2	2.6	3.1	4.0	2.3	208

levels and 0 if they fell by more than 20 percent. Columns 4–6 report probit estimates using the same set of regressors. The cone indicator again remains the most significant predictor across specifications.⁵³

Taken together, this evidence indicates that the persistent heterogeneity in postpandemic commuting patterns is well accounted for by the coordination mechanism implied by our model. Because this coordination occurs partly outside firm boundaries, individual firms' return-to-office policies may be insufficient to restore the prepandemic equilibrium. City-level interventions—such as taxing remote work or subsidizing commuting—could in principle do so, but their welfare implications and effectiveness remain open questions that we explore next.

VI. Welfare

The model gives us an expression to compute the average stationary equilibrium welfare in a city as a function of the measure of commuters. Evaluating equation (8) in a stationary equilibrium and rearranging terms, the average welfare $\mathcal{W}_j$ can be computed as

$$(35) \quad \mathcal{W}(L_{cj}^{ss}) = \left(\frac{L_{cj}^{ss}}{L_j} \right) V_c(L_{cj}^{ss}) + \left(1 - \frac{L_{cj}^{ss}}{L_j} \right) V_m(L_{cj}^{ss}).$$

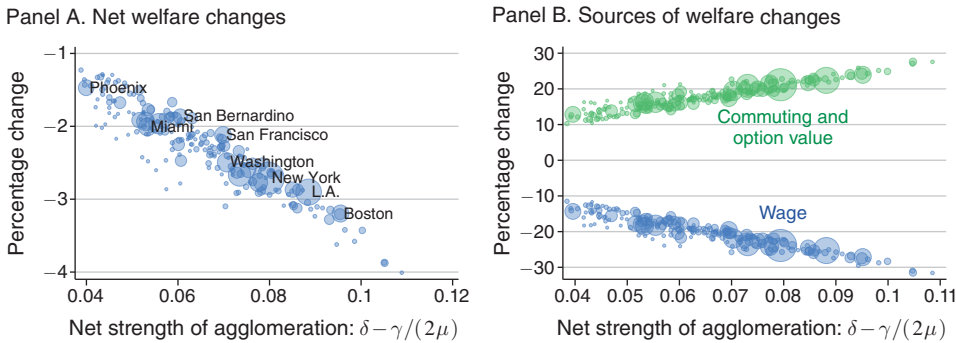
This is a utilitarian welfare function where all workers in the same delivery mode receive the same utility, despite living at different distances from the CBD.⁵⁴ In a city with multiple stationary equilibria, this welfare varies across stationary equilibrium. For the 208 cities inside the cone, we can then compute $\hat{\mathcal{W}}_j \equiv 100 \times (\mathcal{W}(L_{cj}^{ss, low}) / \mathcal{W}(L_{cj}^{ss, high}) - 1)$ as the percentage change between the welfare in the low-commuting versus the high-commuting stationary equilibrium. All cities in the cone would lose from being permanently pushed to the low-commuting stationary equilibrium.⁵⁵ Table 3 shows some moments of the distribution of welfare losses.⁵⁶

⁵³ Supplemental Appendix D.D3 shows similar results under an alternative definition of switching.

⁵⁴ This occurs because, within each group, individuals choose distance to trade off rents and commuting costs.

⁵⁵ Although we do not have a proof, the intuitive reason why all cities in the multiplicity cone would rather be in the commuting-intensive stationary equilibrium is that the only active externality in our quantification is a positive agglomeration externality. This implies that, in all equilibria, there is too little commuting compared to the efficient benchmark.

⁵⁶ Table D2 in the Supplemental Appendix shows actual welfare changes for the largest US cities in our calibration sample.

FIGURE 12. WELFARE RATIO $\hat{\mathcal{W}}$ AND ITS DECOMPOSITION

Notes: The left panel shows a scatterplot between $\hat{\mathcal{W}}$ and the net strength of agglomeration externalities, $\delta - \gamma/(2\mu)$. The right panel shows the percentage difference in wages across equilibria, $\hat{w}$, and the residual commuting and option value gains χ , plotted against the net strength of agglomeration externalities, $\delta - \gamma/(2\mu)$. The marker size is proportional to total city employment. These scatterplots can only be computed for the set of cities in the cone of multiplicity.

The median potential loss from such a switch to the low-commuting stationary equilibrium is 2.2 percent, with a range from 1.2 percent to 4.0 percent.⁵⁷ The loss stems from a drop in average wages, which is only partially compensated by a reduction in commuting costs and changes in the option values. A city switching equilibrium has lower wages both for commuters and remote workers since each individual interacts with a smaller average number of agents, $\tilde{L}_{cj}$. Fewer interactions lead to diminished production externalities. Specifically, the change in average wages is given by

$$(36) \quad \hat{w}_j \equiv 100 \times \left(\frac{\left(\frac{L_{cj}^{ss, low}}{L_j} \right) w_{cj, low} + \left(1 - \frac{L_{cj}^{ss, low}}{L_j} \right) w_{mj, low}}{\left(\frac{L_{cj}^{ss, high}}{L_j} \right) w_{cj, high} + \left(1 - \frac{L_{cj}^{ss, high}}{L_j} \right) w_{mj, high}} - 1 \right).$$

The residual component of the welfare impact that accounts for changes in commuting costs and option values can be constructed as $\chi_j \equiv \hat{\mathcal{W}}_j - \hat{w}_j$. Figure 12, panel B plots $\hat{w}_j$ and χ_j against the net strength of agglomeration externalities, $\delta_j - \gamma_j/(2\mu)$, for all cities estimated to be in the cone. The fall in average wages is quite severe, between 15 percent and 35 percent. This large magnitude is to be expected, however, because agglomeration externalities tend to be significant precisely for cities more likely to be in the multiplicity region. Net welfare losses are then muted by the combined effect of changes in the average commuting disutility and option values. Overall, the net strength of the agglomeration externality (Figure 12, panel A) is a strong predictor of the final welfare loss; a regression of the estimated loss on $\delta_j - \gamma_j/(2\mu)$ delivers an R^2 of 0.85.

⁵⁷In Supplemental Appendix D.D5 we recompute this table for different values of $\xi - \theta$. Supplemental Appendix Table D7 shows that, relative to a benchmark average loss of 2.3 percent, varying $\xi - \theta$ in the range $[-0.03, 0.03]$ implies a mean loss between 1.6 percent and 3.3 percent.

In Figure D2 in the Supplemental Appendix, we also compare welfare losses to the observed drop in trips to the CBD. The largest drops in CBD trips are concentrated around welfare losses of 2.5 to 3 percent. In particular, we find that the welfare loss of the average resident in cities with trips to the CBD of 60 percent or less of the prepandemic level is 2.7 percent.⁵⁸

VII. Conclusion

We have proposed a theory of remote work and city structure. At its core, our theory features a coordination problem: Workers prefer to commute if others do as well and prefer to work from home otherwise. This mechanism can lead to multiplicity in stationary equilibria if agglomeration forces are sufficiently larger than congestion forces and remote-work technology is decent relative to in-office technology. In the resulting dynamic discrete choice model, large exogenous changes in the number of commuters, a state variable of the model, can put the city on an equilibrium path that makes it converge to an alternative stationary equilibrium. We have argued that the COVID-19 pandemic constituted such an exogenous shock and that, as a result, many large cities have switched from a stationary equilibrium where most workers commuted to one in which there is a large share of agents working from home. Small cities, in contrast, tend not to exhibit this multiplicity and therefore have converged back to the prepandemic stationary equilibrium. We find consistent evidence from our estimated model for 278 US cities, from cell phone-based individual mobility data, and from house price distance gradients. An arguably uniform shock to the labor delivery mode across cities has revealed which cities were exposed to this coordination problem—and thus, presented multiple stationary equilibria—before the pandemic, a rather unique empirical event.

The model can be used to estimate the welfare losses associated with the switch to remote work for all cities that exhibit this type of multiplicity in stationary equilibria. We find that welfare in the commuting-intensive equilibrium is between 1.2 percent and 4.0 percent larger than in the remote-work-intensive equilibrium, a significant, but not overwhelming, cost. This modest cost is, however, the result of a large drop in wages compensated by substantial utility gains from option values and commuting costs.

Although our analysis can fit the available data quite well, it inevitably abstracts from a number of important forces and issues. First, our model studies cities in isolation but not as a system of cities with migration, trade, and investment links. This is potentially relevant since the spatial equilibrium condition across cities will determine local welfare effects and population flows. The welfare implications we estimate for some of the large US cities appear to be partially translating into smaller populations. Similarly, some of the small cities with unique stationary equilibria might grow and gain industries that transform them in ways that could put them in the multiplicity cone. Of course, remote workers could also commute across

⁵⁸ As welfare falls, one might also expect a population response. In Supplemental Appendix B.B3 we find that large cities are indeed experiencing a small differential drop in residents, although the changes are rather small and not very correlated with the change in trips to the CBD (see Supplemental Appendix Figure B4b). This pattern is probably due to a low migration elasticity across cities, large migration costs, and the myriad of idiosyncratic shocks affecting individual cities. This is the reason we do not consider between-city employment flows in our analysis.

cities, another issue we abstract from but that has received some recent attention in the media. Second, although we model remote work as “partial remote work,” where agents work from home only a fraction μ of days, we have kept this fraction constant rather than the agent’s choice. Of course, agents in different cities, or in different parts of a city, or with different occupations or skills might choose different shares. Understanding the importance of these choices and the resulting heterogeneity across individuals could be important but complicates the analysis significantly. Third, some of the spillovers between workers could be internalized by firms and cities, which might then proactively affect workers’ delivery mode choice. Understanding the extent to which active policy can determine the dynamic path and stationary equilibrium that a city converges to is an interesting topic for future research.

The repercussions of remote work are intriguing and potentially transforming for the organization of work in cities, firms, and labor markets. Our paper hopes to contribute to what will surely be an active area of research in the future, as the organization of work changes to accommodate the booming phenomenon of remote work.

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